

Common Armament Tester for Fighter (CAT-F)

DRAFT STATEMENT OF OBJECTIVES (SOO)

Prepared by:

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Purpose The Government requires a common armament tester to execute both flightline and backshop requirements for United States Air Force (USAF) fighter fleets. The Government's primary objective is development, manufacture, deployment, and sustainment of a Common Armament Tester for Fighter (CAT-F). The CAT-F will be used at both Organizational (O-LVL) and Intermediate (I-LVL) Level maintenance. The tester shall be able to execute preload, clean wing, and fault isolation armament requirements and replace, or augment, legacy testers for the A-10, F-15, F-16. The CAT-F shall be cyber secure, nuclear certified and use a Modular Open System Architecture approach. The program will require industry to utilize a CAT-F specific Government Reference Architecture (GRA). The CAT-F acquisition will also be a solution for Diminishing Manufacturing Sources and Materiel Shortages (DMSMS) and Obsolescence Issues across several legacy testers.

The Government is providing industry with the overall solicitation objectives which allow flexibility for each offeror to propose an innovative development approach to achieve the Government's CAT-F requirements. Offerors shall use the Statement of Objectives (SOO) and attachments with the Request for Proposal (RFP) as a basis for preparing their proposal. Proposals shall include a contractor Statement of Work (SOW) and expected Contract Data Requirements Lists (CDRLs). Proposals shall also include an Integrated Master Schedule (IMS) that indicates their schedule to design, manufacture, and deliver a CAT-F solution for the F-16, F-15, and A-10 platforms.

CAT-F Scope

The "CAT-F System" is defined as an O-Level (Flightline) and I-Level (Backshop) tester. The CAT-F System will be comprised of components with each component requiring kitting designed for safe storage and rapid deployment.

The required components are:

- O-Level Handheld tester
- Aircraft Specific O-Level Cable and Adapter Kit
- I-Level Functional and Fault Isolation tester
- Aircraft Specific I-Level Cable and Adapter Kit

Statement of Objectives – Common Armament Tester for Fighter (CAT-F)

O-Level Handheld Tester

The CAT-F O-Level Handheld tester must be able to execute legacy O-Level Maintenance (Flightline) Pre-load and Clean Wing requirements for the F-16, F-15, and A-10 fighter aircraft. The Handheld Tester must work independently to execute O-Level test requirements and store testing results. The test shall be compatible with I-Level Functional and Fault Isolation Tester.

Aircraft Specific O-Level Cable and Adapter Kit

The Cable/Adapter TPS Kit is expected to be aircraft specific and contain all-appropriate cables and adapters needed to execute pre-load and clean wing testing required at O-Level Maintenance.

I-Level Functional and Fault Isolation Tester

The CAT-F I-Level Functional and Fault Isolation Tester must be able to execute legacy I-Level (Backshop) requirements for the F-16, F-15, and A-10 fighter aircraft and shall incorporate the O-Level Handheld tester in its design.

Aircraft Specific I-Level Cable and Adapter Kit

The Cable/Adapter TPS Kit is expected to be aircraft specific and contain the appropriate cables and adapters needed to execute Fault Isolation requirements at I-Level Maintenance.

CAT-F Component Quantities

The expected component quantities listed are based on current fleet status and may be adjusted to meet future Force Structure Changes. This will be an Indefinite Delivery, Indefinite Quantity contract (IDIQ). Table 1.0 indicates the Governments estimated component quantities.

Table 1.0 CAT-F Estimated Component Quantities

CAT-F Component	F-16 with Spares	F-15 with Spares	A-10 with Spares
O-Level Handheld Testers	304	172	68
O-Level Cable and Adapter Kits	304	172	68
I-Level Functional/Fault Isolation Testers	143	80	42
I-Level Cable and Adapter Kits	143	80	42

Product Support / Logistics Considerations

- Solution considers COTS and equipment that will expedite development, production, and fielding
- Solution that reduces legacy equipment footprint and able to deploy worldwide
- Provide equipment solutions that deliver a reliable option and minimizes the potential for DMSMS
- Describe the MOSA approach to deliver CAT-F capability
- IP considerations and technical data package.

Statement of Objectives – Common Armament Tester for Fighter (CAT-F)

Period of Performance (PoP)

The period of performance will be a one (1) year Basic with four (4) one-year Options.

Contractor Statement of Work (SOW)

Offeror shall develop a SOW using the SOW template (SOO Attachment 5). The SOW must clearly indicate the work to complete the CAT-F with detailed tasks, technical requirements, evaluation and exit criteria, and recommended program deliverables. The SOW may be incorporated into the resulting award and should not include any proprietary or sensitive company information.

Integrated Master Schedule (IMS)

The offerors are expected to develop and provide a time-based Integrated Master Schedule (IMS) detailing the tasks, key milestones, and delivery dates necessary to ensure successful program/contract execution. The IMS should be traceable to the offer's developed Contractor SOW. The Offeror's submitted IMS shall address all 5 contract years.

STATEMENT OF OBJECTIVES (SOO) Attachments:

Attachment 1: SYSTEM REQUIREMENT DOCUMENT (SRD)

Attachment 2: F-16/F-15/A-10 ALTERNATE MISSION EQUIPMENT (AME)

Attachment 3: CAT-F AME MAINTENANCE LEVEL REQUIREMENTS

Attachment 4: CAPABILITY DEVELOPMENT DOCUMENT (CDD) SUMMARY

Attachment 5: PREPARATION OF STATEMENT OF WORK (SOW)

ATTACHMENT 1:

Common Armament Tester for Fighter (CAT-F)

DRAFT SYSTEM REQUIREMENT DOCUMENT (SRD)

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15 July 2024

The Common Armament Tester for Fighter (CAT-F) System Requirement Document (SRD) defines the system-level functional and performance requirements for a common armament tester for multiple fighter Aircraft. The SRD is derived from the CAT-F Capability Development Document (CDD) and Concept of Operations (CONOPS), system-level performance metrics.

GRA Ref.	Name	Text/Description	Verification
SRD_1.0	CAT-F System Requirements (New Development)		
SRD_1.1	Required States and Modes	The Testers non-operational and operational modes are defined below: a. Non-Operational modes including the tester's system storage and system transportation, if applicable. b. Operational modes including Tester Maintenance and UUT/DUT Testing. The Tester maintenance modes consists of System Administration activities supporting software updates and approved user access, Tester Self-Test and Tester Calibration.	Demonstration
SRD_1.1.1	Tester Self-Test/Built-In Test	The Tester Self-Test/Built in Test (BIT) shall be designed in accordance with MIL-PRF-32070. If faults are discovered, the core tester's functional status shall be displayed to the user and shall not require the use of external technical manuals to determine functional status.	Analysis
SRD_1.1.1.1	Power-On Confidence Test	The Tester shall provide a Power-On Confidence Test to ensure proper operation prior to initiation/stimulation of UUT test. The Power-On Confidence Test shall be automatically performed upon system power-up and shall include running the malware detection defined in 3.8.1.6 of this specification. Power-On Confidence test shall be completed in 10 minutes or less (T) with an objective of 2 minutes (O).	Analysis Demonstration
SRD_1.1.1.2	Self-Test	The Tester shall have an integrated Self-Test capability for failure detection and fault isolation of all system components, and shall provide indications of out of range conditions of the tester. The Self-Test shall ensure that stimulus and measurement devices are operating within tolerance. Self-Test shall run upon system start up and when manually commanded by the user. Self-Test during TPS execution shall not be enabled.	Demonstration
SRD_1.1.1.2.1	Self-Test Data	The Tester on-board fault and Self-Test data shall be collected, stored, and provided IAW IEEE 1636, IEEE 1636.1, and IEEE 1636.2 compatible formats to maintenance personnel during and after maintenance operations.	Demonstration
SRD_1.1.1.2.2	Self-Test Execution Time	The Tester Self-Test execution time shall be no longer than 10 minutes (T) with an objective of 1 minute (O).	Demonstration
SRD_1.1.1.2.3	Self-Test Fault Detection	The Tester Self-Test shall detect 100 percent of detectable faults. A detectable fault is a failure of a Tester Replaceable Unit (TRU) that is exhibited at the interface of the TRU and is outside the defined performance parameter or threshold for that particular function.	Demonstration
SRD_1.1.1.2.4	Self-Test Fault Isolation Rate – 1 Component	The Tester Self-Test shall isolate at least 90% of detected faults to a single system component (e.g.: tester unit, interface unit, cable number/connector number/pin number, software, etc.)	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.1.1.2.5	Self-Test Fault Isolation Rate – 2 Components	The Tester Self-Test shall isolate at least 95% of detected faults to two system components (e.g.: tester unit, interface unit, cable number/connector number/pin number, software, etc.)	Demonstration
SRD_1.1.1.2.6	Self-Test Fault Isolation Rate – 3 Components	The Tester Self-Test shall isolate 100% of detected faults to three or less system components (e.g.: tester unit, interface unit, cable number/connector number/pin number, software, etc.).	Demonstration
SRD_1.1.2	Test Program Set Requirements	All armament UUT TPSs developed for use by the Tester shall be designed IAW MIL-PRF-32070.	Analysis
SRD_1.1.2.1	Unit Under Test Data Log and Documentation	The Tester software shall collect, store, and provide upon command, the following data in compliance with IEEE-STD-1636.1 for future export: General Information UUT Serial Number - Installed LRU serial numbers if applicable (A UUT for test purposes may contain other subassemblies which are themselves UUTs and separately serial numbered) - Tester serial number - Operator and date - Computer Program Identification Number (CPIN) - Checksum/Signature of CPIN to indicate Software unaltered - Tester Self-Test Status - Total Test Time - Overall UUT status (Pass/Fail) For each test sequence in the TPS - Upper tolerance value - Measured value - Lower tolerance value - Expected digital data - Received digital data word, in both actual state and in “layman’s terms” - Power Quality Information in accordance with paragraph 3.3.3.7. - Time Stamp - Indictments (or callouts) determined through fault isolation to include Failed Connector and Pin and input Connector/Pin(s). Information shall be traceable to the test program. - Number of times an individual test sequence was run The Tester shall retain the UUT test data log after an incomplete test or test termination.	Analysis

GRA Ref.	Name	Text/Description	Verification
SRD_1.1.2.2	Real Time Display of Test Results and Status	The Tester shall provide real-time test status to the operator: a. UUT Serial Number b. CPIN Number c. Test Number being executed d. Upper tolerance value e. Measured value f. Lower tolerance value g. 1553 status h. External Elapsed Time Indicator i. Expected digital data j. Received digital data in both actual state and in “layman’s terms” k. Recommended Failure Modes e.g. call out most common LRU, Cable etc. Failed Connector and Pin and input Connector/Pin(s) l. Measurement values for loop back tests and adjustments	Analysis
SRD_1.1.2.3	End of Test Report	At the conclusion of TPS execution, the Tester shall display and log a test report for indicating the pass/fail status and certifying the functionality of the UUT.	Analysis
SRD_1.1.2.4	Safe-To-Turn-On Test	The UUT TPSs shall perform safe-to-turn-on tests prior to executing the TPS IAW MIL-PRF-32070A.	Analysis
SRD_1.1.2.5	Test Sequencing	The UUT TPSs shall automatically sequence through all test steps pausing only when operator input is required, when operator manually aborts test sequence, when a failure is detected, or until all test steps are successfully completed. Upon failure of a test step or the operator manually aborting the test sequence, the Tester shall display the fault data and allow the operator to retest the failed step, continue testing, or discontinue testing.	Analysis
SRD_1.1.2.6	Test Termination	The UUT TPSs shall provide a termination feature to perform an orderly termination of a test in progress (i.e., terminate external interfaces) leaving both the Tester and UUT in a safe state.	Analysis
SRD_1.1.2.7	Test Entry Points	The Tester shall allow the operator to select and run test sequences from operator selectable entry points. At a minimum, entry points shall provide autonomous groups of tests based on UUT functionality. Test Entry Points shall provide the capability for the operator to select and execute testing of specific functional areas of the UUT individually and separate from other functional areas, as well as complete end-to-end testing. UUT certification shall require an end-to-end go chain.	Analysis
SRD_1.1.2.8	Manual Test Sequencing	Manual test sequencing shall perform one test step in the sequence and pause for the operator to initiate the next step.	Analysis
SRD_1.1.2.9	Test Control Selection	Selection between automatic and manual test control shall be available, except when prohibited due to sequencing and/or timing requirements for tests.	Analysis
SRD_1.1.2.10	Test Data Transfer	The Tester shall provide a cyber-secure compliant capability to transfer test data to an approved external device through an existing secure data port when commanded by the administrator.	Analysis

GRA Ref.	Name	Text/Description	Verification
SRD_1.1.2.11	Clearing Test Data	The Tester shall have the capability to clear non-volatile memory and test data when commanded by an administrator.	Analysis
SRD_1.1.2.12	Pre-Test External Cable Identification Cable and Cable Testing	The Tester shall verify the external cable configuration/hook-up prior to the start of any test. The Tester shall notify operator of any incorrect cable configurations prior to the start of any test. The Tester shall prevent test initiation when cables are incorrectly configured. The Tester shall test all Test Program Hardware to include Interface Test Adapters (ITA) and ITA cables as part of the UUT TPS IAW MIL-PRF-32070A paragraph 3.3.6.	Analysis
SRD_1.1.2.13	Stray Voltage Test	The Tester shall be capable of detecting stray voltage, as required prior to munitions load.	Analysis
SRD_1.1.2.14	Unit Under Test Fault Detection	The Tester shall detect 100 percent of detectable faults of system LRU failures, based on the available data offered by on-aircraft or off-aircraft equipment. A detectable fault is a failure of the UUT that is exhibited at the interface of the UUT and is outside the defined performance parameter or threshold for that particular function.	Analysis
SRD_1.1.2.15	Unit Under Test Cooling	The Tester shall provide a means to interface with UUT cooling carts when required.	Analysis
SRD_1.1.2.16	Interface Test Adapters	TPS ITAs shall not utilize active circuit components without prior Government approval.	Analysis
SRD_1.1.2.17	Functional and Parametric Test Data	Functional and Parametric data for both instruments and test stations should be formatted IAW ATML standards specified in IEEE 1671.2, IEEE 1871.1, as well as IEEE 1671.6 and 1871.2 respectively.	Analysis
SRD_1.1.2.18	Master Conformance Index	The Master Conformance Index present within the Tester shall be developed IAW IEEE 1671.4 ATML Test Configuration.	Analysis
SRD_1.1.2.19	Receiver Adapter Interface	The Receiver Adapter Interface present within the Tester shall be developed IAW IEEE 1641 Annex K.	Analysis
SRD_1.1.2.20	UUT Test Requirements and Product Design Data	UUT Test Requirements and Product Design Data shall be defined and designed respectively IAW IEEE 1671.1 ATML Test Description, as well as ANSI/EIA 682, Electronic Design Interchange Format (EDIF), and IEEE 1671.3 ATML UUT Description.	Analysis
SRD_1.1.2.21	Digital Test Interchange Format	Interchangeable data present within the Tester shall be formatted IAW IEEE 1445 Digital Test Interchange Format (DTIF).	Analysis
SRD_1.1.2.22	Test Program Documentation	All documentation referring to the Test Programs shall be developed IAW IEEE P1671.1 ATML Test Description, and IEEE 1671 Annex C.2 ATML Wire list Schema.	Analysis
SRD_1.1.2.23	Design for Testability	Designs for Testability will be created IAW IEEE 1149.1 Test Access Port and Boundary-Scan Architecture, IEEE 1149.4 Mixed Signal Test Bus, and IEEE 1149.6 Boundary-Scan Testing of Advanced Digital Networks.	Analysis

GRA Ref.	Name	Text/Description	Verification
SRD_1.1.2.24	Adapter Functional and Parametric Data	Adapter Functional and Parametric Data shall be IEEE 1671.5 ATML Test Adapter, and IEEE 1871.2 Recommended Practice for Intrinsic Signal Path Information.	Analysis
SRD_1.1.2.25	Test Duration	(U) Total test time using the Common Handheld Tester and the Preload TPS/Kit (O-level MDS specific) shall be less than the current duration of all comparable operational and/or functional testing. The O-LVL 180-day testing time will not exceed 6 Hours and all testing at I-LVL will be under 1 hour per AME.	Test
SRD_1.1.2.26	Software Developer's Guide	The contractor shall develop the TPS style guide IAW ATS standard	Inspection
SRD_1.2	Legacy Testing	(U) The CAT-F will be capable of performing all current legacy armament system tests and meet required nuclear certification. The CAT-F will replace multiple current USAF fighter Aircraft armament testers and shall execute all legacy inspection and troubleshooting requirements. The tester will be required to conduct armament preload and fault isolation testing on active model versions of the F-16 and F-15 fleets. This tester should be considered as a replacement armament tester for the A-10 fleet and be a future consideration for newer generation fighters. To verify the requirement, the CAT-F will be tested in all operationally representative environments and on all appropriate fighter Aircraft. The results will be compared to the appropriate legacy tester.	Test
SRD_1.2.1	A-10 AME	The tester shall be able to execute all A-10 AME requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.1	Stations 1 and 11 Pylon (with MAU-50 installed)	The tester shall be able to execute all Stations 1 and 11 Pylon (with MAU-50 installed) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.2	Stations 2 and 10 Pylon (with MAU-50 installed)	The tester shall be able to execute all Stations 2 and 10 Pylon (with MAU-50 installed) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.3	Stations 3 & 9 Pylon (with MAU-40 installed)	The tester shall be able to execute all Stations 3 & 9 Pylon (with MAU-40 installed) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.4	Stations 4 & 8 Pylon (with MAU-40 installed)	The tester shall be able to execute all Stations 4 & 8 Pylon (with MAU-40 installed) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.5	Station 5 Pylon (with MAU-40 installed)	The tester shall be able to execute all Station 5 Pylon (with MAU-40 installed) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.6	Station 6 Pylon (with MAU-40 installed)	The tester shall be able to execute all Station 6 Pylon (with MAU-40 installed) requirements I.A.W. the specific MDS Technical Orders.	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.2.1.7	Station 7 Pylon (with MAU-40 installed)	The tester shall be able to execute all Station 7 Pylon (with MAU-40 installed) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.8	TER-9/A or Variant installed	The tester shall be able to execute all TER-9/A or Variant installed requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.9	Dual Rail Adaptor (DRA) with LAU-105s and LAU-128 installed	The tester shall be able to execute all Dual Rail Adaptor (DRA) with LAU-105s and GMU installed requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.10	LAU-131	The tester shall be able to execute all LAU-131 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.11	Chaff Flair	The tester shall be able to execute all Chaff Flair requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.12	LAU-117	The tester shall be able to execute all LAU-117 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.13	BRU-61	The tester shall be able to execute all BRU-61 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.14	LAU-88	The tester shall be able to execute all LAU-88 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.15	Gun Control Unit	The tester shall be able to execute all Gun Control Unit requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.16	Pylon 5CU-A/SCU-8	The tester shall be able to execute all Pylon 5CU-A/SCU-8 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.17	AN/ALE-47 CMDS	The tester shall be able to execute all AN/ALE-47 CMDS requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.1.18	LAU-128	The tester shall be able to execute all LAU-128 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2	F-15 AME	The tester shall be able to execute all F-15 AME requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.1	PYLON	The tester shall be able to execute all PYLON requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.2	BRU-46/47	The tester shall be able to execute all BRU-46/47 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.3	BRU-61	The tester shall be able to execute all BRU-61 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.4	CBU-103105,107,118 System Check	The tester shall be able to execute all CBU-103105,107,118 System Check requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.5	GBU-31 (JDAM) System Check	The tester shall be able to execute all GBU-31 (JDAM) System Check requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.6	GUB-31B UAi System Check	The tester shall be able to execute all GUB-31B UAi System Check requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.7	M-61A1	The tester shall be able to execute all M-61A1 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.8	Alm-9M Missile System	The tester shall be able to execute all Alm-9M Missile System requirements I.A.W. the specific MDS Technical Orders.	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.2.2.9	GBU-38 UAI/GBU-54 UAI System Check	The tester shall be able to execute all GBU-38 UAI/GBU-54 UAI System Check requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.10	AGM-158 UAI System Check	The tester shall be able to execute all AGM-158 UAI System Check requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.11	AIM-9X/X-2 Missile System	The tester shall be able to execute all AIM-9X/X-2 Missile System requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.12	AIM-120 LAU-106 Missile Systems	The tester shall be able to execute all AIM-120 LAU-106 Missile Systems requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.13	AIM-120 LAU-128 Missile System	The tester shall be able to execute all AIM-120 LAU-128 Missile System requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.14	Jettison Checks Without Fuel Tanks	The tester shall be able to execute all Jettison Checks Without Fuel Tanks requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.15	Jettison Checks with Fuel Tanks	The tester shall be able to execute all Jettison Checks with Fuel Tanks requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.16	BRU-46/47 Single Carriage (Pylon)	The tester shall be able to execute all BRU-46/47 Single Carriage (Pylon) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.17	BRU-47	The tester shall be able to execute all BRU-47 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.18	SUU-20	The tester shall be able to execute all SUU-20 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.19	MAU-12	The tester shall be able to execute all MAU-12 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.20	LAU-128 (AIM-9L/M)	The tester shall be able to execute all LAU-128 (AIM-9L/M) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.21	LAU-128 (AIM-120)	The tester shall be able to execute all LAU-128 (AIM-120) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.22	LAU-120 (AIM-120)	The tester shall be able to execute all LAU-120 (AIM-120) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.2.23	GBU-39 UAI System Check	The tester shall be able to execute all GBU-39 UAI System Check requirements I.A.W. the specific MDS Technical orders.	Demonstration
SRD_1.2.2.24	GBU-53 UAI System Check	The tester shall be able to execute all GBU-53 UAI System Check requirements I.A.W. the specific MDS Technical orders.	Demonstration
SRD_1.2.2.25	Multipurpose Booster (MPB) UAI System Check	The tester shall be able to execute all MPB UAI System Check requirements I.A.W. the specific MDS Technical orders.	Demonstration
SRD_1.2.2.26	ALE-58A System Check	The tester shall be able to execute all ALE-58A System Check requirements I.A.W. the specific MDS Technical orders.	Demonstration
SRD_1.2.2.27	SUU-60B/A	The tester shall be able to execute all SUU-60B/A requirements I.A.W. the specific MDS Technical orders.	Demonstration
SRD_1.2.2.28	SUU-59B/A	The tester shall be able to execute all SUU-59B/A requirements I.A.W. the specific MDS Technical orders.	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.2.2.29	SUU-59C/A	The tester shall be able to execute all SUU-59C/A requirements I.A.W. the specific MDS Technical orders.	Demonstration
SRD_1.2.2.30	SUU-73/A	The tester shall be able to execute all SUU-73/A System Check requirements I.A.W. the specific MDS Technical orders.	Demonstration
SRD_1.2.2.31	Missile Launcher Auxiliary Relay Panel Check	The tester shall be able to execute all Missile Launcher Auxiliary Relay Panel Check requirements I.A.W. the specific MDS Technical orders.	Demonstration
SRD_1.2.2.32	IRST Pod System Check	The tester shall be able to execute all IRST Pod System Check requirements I.A.W. the specific MDS Technical orders.	Demonstration
SRD_1.2.2.33	B61 UAI System Check	The tester shall be able to execute all B-61 UAI System Check requirements I.A.W. the specific MDS Technical orders.	Demonstration
SRD_1.2.3	F-16 AMU	The tester shall be able to execute all F-16 AME requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.1	Preload Functional Checks, 7560 Replacement Tess, Pre-Block Aircraft	The tester shall be able to execute all Preload Functional Checks, 7560 Replacement Tess, Pre-Block Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.1.1	MAU--12 on Aircraft	The tester shall be able to execute all MAU--12 on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.1.2	External Fuel Tank Pylon on Aircraft	The tester shall be able to execute all External Fuel Tank Pylon on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.1.3	AIM 9 Missile Launcher on Aircraft	The tester shall be able to execute all AIM 9 Missile Launcher on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.1.4	LAU-129 Missile Launcher on Aircraft	The tester shall be able to execute all LAU-129 Missile Launcher on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.1.5	TER-9A on Aircraft	The tester shall be able to execute all TER-9A on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.1.6	On Aircraft Launcher Umbilical Only	The tester shall be able to execute all On Aircraft Launcher Umbilical Only requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.1.7	SUU-20 on Aircraft	The tester shall be able to execute all SUU-20 on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.1.8	M61A1 20MM Gun System Control Circuits / Gun Control Unit (GCU)	The tester shall be able to execute all M61A1 20MM Gun System Control Circuits / Gun Control Unit (GCU) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.1.9	On Aircraft WWP 1760 Umbilical Only	The tester shall be able to execute all On Aircraft WWP 1760 Umbilical Only requirements I.A.W. the specific MDS Technical Orders.	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.2.8.1.10	SCRIU Equipped WWP	The tester shall be able to execute all SCRIU Equipped WWP requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.1.11	BRU-57 on Aircraft	The tester shall be able to execute all BRU-57 on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2	Preload Functional Checks-75060 Replacement Tests, Post-Block Aircraft	The tester shall be able to execute all Preload Functional Checks-75060 Replacement Tests, Post-Block Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.1	MAU-12 on Aircraft	The tester shall be able to execute all MAU-12 on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.2	External Fuel Tank Pylon on Aircraft	The tester shall be able to execute all External Fuel Tank Pylon on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.3	TER-9A on Aircraft	The tester shall be able to execute all TER-9A on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.4	AIM-9 Missile Launcher on Aircraft	The tester shall be able to execute all AIM-9 Missile Launcher on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.5	LAU-129 Missile Launcher on Aircraft	The tester shall be able to execute all LAU-129 Missile Launcher on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.6	LAU-118 Missile Launcher on Aircraft	The tester shall be able to execute all LAU-118 Missile Launcher on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.7	On Aircraft Launcher Umbilical Only LAU- 117	The tester shall be able to execute all On Aircraft Launcher Umbilical Only LAU-117 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.8	On Aircraft Launcher Umbilical only LAU 118 Launcher	The tester shall be able to execute all On Aircraft Launcher Umbilical only LAU 118 Launcher requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.9	On Aircraft Rocket Launcher Umbilical only	The tester shall be able to execute all On Aircraft Rocket Launcher Umbilical only requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.10	SUU-20 on Aircraft	The tester shall be able to execute all SUU-20 on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.11	M61A1 20MM Gun System Control Circuits / Gun Control Unit (GCU)	The tester shall be able to execute all M61A1 20MM Gun System Control Circuits / Gun Control Unit (GCU) requirements I.A.W. the specific MDS Technical Orders.	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.2.8.2.12	On Aircraft WWP, 1760 Umbilical Only	The tester shall be able to execute all On Aircraft WWP, 1760 Umbilical Only requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.13	BRU-57 on Aircraft	The tester shall be able to execute all BRU-57 on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.14	Under Wing Adapter	The tester shall be able to execute all Under Wing Adapter requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.15	ALE-50 Pylon on Aircraft	The tester shall be able to execute all ALE-50 Pylon on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.16	BRU-46 on Aircraft	The tester shall be able to execute all BRU-46 on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.17	Left Hardpoint Interface	The tester shall be able to execute all Left Hardpoint Interface requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.2.18	Right Hardpoint Interface	The tester shall be able to execute all Right Hardpoint Interface on Aircraft requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3	O Level Stores System Tester (SST) - Replacement Tests	The tester shall be able to execute all O Level Stores System Tester (SST) - Replacement Tests requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.1	Aircraft station 1, 2, 3 a, 8 and 9 interface connection circuits	The tester shall be able to execute all Aircraft station 1, 2, 3a, 8 and 9 interface connection circuits requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.2	Aircraft station 3, 4, 6 and 7 interface connection circuits	The tester shall be able to execute all Aircraft station 3, 4, 6 and 7 interface connection circuits requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.3	Aircraft station 5 (centerline station) interface connection circuits Pre ASQ-236 Modifications	The tester shall be able to execute all Aircraft station 5 (centerline station) interface connection circuits Pre ASQ-236 Modifications requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.4	Aircraft station 5 (centerline station) interface connection circuits Post ASQ-236 Modifications	The tester shall be able to execute all Aircraft station 5 (centerline station) interface connection circuits Post ASQ-236 Modifications requirements I.A.W. the specific MDS Technical Orders.	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.2.8.3.5	ASQ-236 Centerline Pylon on Aircraft, station 5	The tester shall be able to execute all ASQ-236 Centerline Pylon on Aircraft, station 5 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.6	Centerline Pylon on Aircraft, station 5	The tester shall be able to execute all Centerline Pylon on Aircraft, station 5 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.7	AIM - 9 Launcher on Aircraft, on stations 1, 2,3A, 7A, 8, 9	The tester shall be able to execute all AIM - 9 Launcher on Aircraft, on stations 1, 2,3A, 7A, 8, 9 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.8	LAU-129 Launcher on Aircraft, on stations 1, 2, 3A, 7A, 8, 9	The tester shall be able to execute all LAU-129 Launcher on Aircraft, on stations 1, 2, 3A, 7A, 8, 9 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.9	1760 or non-1760 Wing Weapon Pylon Equipped with Special Carry Remote Interface Unit (SCRIU) on Aircraft, stations 3 and 4	The tester shall be able to execute all 1760 or non-1760 Wing Weapon Pylon Equipped with Special Carry Remote Interface Unit (SCRIU) on Aircraft, stations 3 and 4 requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.10	Non - 1760 equipped Wing Weapon Pylon on Aircraft, stations 3, 4, 6, 7 (MIL - STD-1760) W20 Check	The tester shall be able to execute all Non - 1760 equipped Wing Weapon Pylon on Aircraft, stations 3, 4, 6, 7 (MIL - STD-1760) W20 Check requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.11	1760 equipped Wing Weapon Pylon on Aircraft stations 3, 4, 6, 7 (MIL - STD-1760) (W20 Long Check)	The tester shall be able to execute all 1760 equipped Wing Weapon Pylon on Aircraft stations 3, 4, 6, 7 (MIL - STD-1760) (W20 Long Check) requirements I.A.W. the specific MDS Technical Orders.	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.2.8.3.12	1760 equipped Wing Weapon Plon on Aircraft, stations 3, 4, 6, 7 (MIL-STD-1760) (W20 Short Check, 1760 only)	The tester shall be able to execute all 1760 equipped Wing Weapon Plon on Aircraft, stations 3, 4, 6, 7 (MIL-STD-1760) (W20 Short Check, 1760 only) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.13	Aircraft station 5B interface connection circuits (Right Hardpoint Interface Checkout 5R)	The tester shall be able to execute all Aircraft station 5B interface connection circuits (Right Hardpoint Interface Checkout 5R) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.3.14	Aircraft station 5L interface connection circuits (Left Hardpoint interface Checkout 5L)	The tester shall be able to execute all Aircraft station 5L interface connection circuits (Left Hardpoint interface Checkout 5L) requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.4	Troubleshooting (94FI) Checks - 75060 ACPTS and SST	The tester shall be able to execute all Troubleshooting (94FI) Checks - 75060 ACPTS and SST requirements I.A.W. the specific MDS Technical Orders.	Demonstration
SRD_1.2.8.4.1	Gun System Analysis		
SRD_1.2.8.4.2	Multiple Release - Procedures		
SRD_1.2.4	Cables/Adapters	Cables/Adapters will be MDS specific. All Cables and Adapter required to execute testing will be provided in MDS specific kits. Each MDS platform will require cable and adapter kits for the O-Level pre-load, O-Level Fault Isolation, and I-level Fault Isolation testing requirements.	Demonstration
SRD_1.3	Future Armament Testing Requirements	(U) The CAT-F design will include the capability to upgrade by using the Computer Program Identification Number (CPIN) management process to sustain tester upgrades and modification during the tester's life cycle. The developed CAT-F will be built using an open system software architecture that allows for future upgrades and modifications. In addition to software upgrades, the CAT-F shall utilize as much as possible Modular Open Systems Approach (MOSA) design for tester hardware and sustainment. Vendor will be required to submit all requested data to gain government engineering acceptance/approval for the MOSA standard and upgrade capability for software and hardware.	Analysis

GRA Ref.	Name	Text/Description	Verification
SRD_1.4	Nuclear Certification	Nuclear certification for the Tester and applicable nuclear platform TPSs shall be in accordance with AFI 63-125 (Nuclear Certification Program), AFI 91-103, AFI 91-107, AFMAN 91-118, AFMAN 91-119, and MIL-STD-1822.	Analysis
SRD_1.4.1	System Safety	System safety for the Tester shall be conducted IAW the requirements of MIL-STD-882E. The Tester design shall be compliant with requirements defined by AFMAN91-118 and AFMAN91-119.	Analysis
SRD_1.4.1.1	Single Failures	The Tester shall be designed such that no single component failure, common mode failure, human error, or a design feature shall cause a mishap of catastrophic or critical mishap severity categories as defined in MIL-STD-882E.	Analysis
SRD_1.4.1.2	Dual Failures	The Tester shall be designed such that no dual independent component failures, dual independent human errors, or a combination of a component failure and a human error involving safety critical command and control functions, shall cause a mishap of catastrophic or critical mishap severity categories.	Analysis
SRD_1.4.1.3	Design Safety – Critical Functions	The Tester design shall prevent faults in the test equipment or test circuits that could operate critical functions or apply unintended power to the Tester interface.	Analysis
SRD_1.4.1.4	Design Safety – Internal Faults	The Tester design shall prevent internal faults within the tester that could degrade the nuclear or conventional safety of the equipment to be tested.	Analysis
SRD_1.4.1.5	Design Safety – Unintended Signals	The Tester design shall prevent the introduction of signals, voltages, or currents into the Tester system that could degrade nuclear or conventional safety of the equipment to be tested.	Analysis
SRD_1.4.1.6	Design Safety – Unintended Firing	The Tester shall prevent operation or firing of an item under test, except when specifically designed for that purpose.	Analysis
SRD_1.4.1.7	Test Safety – End of Test	The Tester shall ensure any devices, which has been operated during testing, is in the safe or inactivated position when the test ends. A positive indication verifies the safe position of such components.	Analysis
SRD_1.4.1.8	Test Safety – System Failures	The Tester failures or shutdowns shall leave the components under test in a safe or inactivated condition. The Tester shall provide a clear indication of UUT failure status.	Analysis
SRD_1.4.1.9	Electromagnetic Environmental Effects Requirements	The Tester shall be compatible with ordnance classified as Hazards of Electromagnetic Radiation to Ordnance (HERO) SAFE ORDNANCE IAW the requirements of MIL-STD-464C, section 5.9.3, Table 9, as well as MIL-STD-461.	Analysis
SRD_1.5	Systems external interface requirements	The Tester shall accommodate the Unit Under Test (UUT), ancillary equipment, and site interfaces defined below and by reference.	TBD
SRD_1.5.1	Unit Under Test (UUT) external interface description	The UUT interface shall be used to provide AC power, DC power, analog and digital stimuli to the UUTs and measure AC power, DC power, analog and digital signals from the UUTs through the Tester UUT interface.	TBD

GRA Ref.	Name	Text/Description	Verification
SRD_1.5.1.1	UUT communication busses	Specify the communication buses the tester shall provide The following communication interfaces shall be provided by the Tester.	TBD
SRD_1.5.1.1.1	MIL-STD-1553	The MIL-STD-1553 bus interface shall operate as specified in MIL-STD-1553A/B and shall provide an interface for both direct and transformer coupled buses. The following modes of operation shall be supported: Bus controller (BC), Remote terminal (RT), and Bus monitor (BM).	Demonstration
SRD_1.5.1.1.2	MIL-STD-1760		Pull information from AME interface
SRD_1.5.1.2	UUT Discrete Interfaces		
SRD_1.5.2	Power		
SRD_1.5.2.1	Operating Power	List associated voltages (including phases, type of current, and voltage ratings). The Tester shall be capable of operating from 28 VDC and 120/240 VRMS 50 Hz, 60 Hz, and 400 Hz AC (single and multi-phase). Operating power shall comply with the requirements listed in MIL-PRF-28800, MIL-STD-704 and MIL-STD-1332. Automatic and operator initiated shutdown shall provide an orderly, sequential system shutdown to preclude damage to the Automatic Test Equipment (ATE) or UUT and shall not degrade the ATE hardware or software. Operating power for the Tester computer/controller shall have the capability of being isolated from the test power being used to power the UUTs.	Analysis
SRD_1.5.2.1.1	Handheld Tester Power	(U) The CAT-F Handheld Tester will be powered by common design batteries which are non- proprietary, non-rechargeable and replaceable in minimal amount of time. Preload testing will utilize battery power for all go/no-go preload and stray voltage testing. In addition to having the required battery power, user should have the option to use aircraft or Aerospace Ground Equipment (AGE) power for preload testing. Power adapter cables must be provided for all user power options. Verification of handheld preload testing can be accomplished without the use of external power (AGE, Aircraft, building etc.). Verification team will ensure the handheld tester internal batteries can be replaced in under 2 minutes. If required, the verification team will ensure preload testing can easily utilize AGE, Aircraft, and backshop building power to execute requirements.	Demonstration
SRD_1.5.2.1.2	Fault Isolation Tester Power	(U) This equipment shall utilize electric power in accordance with MIL-STD-704 and shall be in accordance with the utilization equipment requirements of MIL-STD-704. As a minimum the primary power source shall operate at 100-240V@400 Hz and 28VDC IAW MIL-STD-704 and include adapters to utilize both aircraft, AGE, and backshop power.	Test

GRA Ref.	Name	Text/Description	Verification
SRD_1.5.2.2	Unit Under Test Power	The Intermediate Tester shall be capable of switching external power as required by the UUT. The Tester shall control on/off switching of power to the UUT as required by the UUT TPS. Operating power for the Tester computer/controller shall have the capability of being isolated from the test power being used to power the UUTs.	Demonstration
SRD_1.5.2.3	Circuit Protection	(U) The CAT-F tester will be developed with built in Circuit protection. The design will ensure the users, CAT-F, and Unit Under Test (UUT) are protected from improper voltage or amperage. Circuit Protection can be obtained by using a traditional fuse designed to 'blow' when a circuit is overloaded and easily replaced or by incorporating a circuit breaker that features a switch designed to trip by improper conditions. The switch then can simply be flip back into the 'on' position to restore power to the tester, providing no additional repairs are required. Verification of KPP2 will be accomplished by applying enough input power to blow fuse or trip circuit breaker in the designed tester, followed by replacing blown fuse or resetting breaker and verify the tester circuits were not damaged. The appropriate sections of MIL-STD-704 (Aircraft Electrical Power Characteristics) will be identified to ensure power requirements for circuit protection are followed.	Test
SRD_1.5.2.4	Power Monitoring	The Intermediate Tester shall measure input power, VRMS for each phase, phase sequence, and frequency information as follows: - As part of the power-up sequence. - At the start of all test sequences. - During each step while the test load is still connected. Report as part of the data for each failure, or for each step if in manual mode. The Tester shall detect and respond to the presence of more than 6 VRMS between the Neutral and Chassis Ground as follows: - If during power-up, notify the operator of a power fault and prevent the Tester power-up sequence. - If during operation, notify the operator and perform an orderly shutdown.	Test
SRD_1.5.2.5	Handheld Power Monitoring	Handheld Tester shall display handheld battery status Tester shall display battery status warning at designated percentages that would impact test execution	TBD
SRD_1.5.3	Grounding		
SRD_1.5.3.1	Controlled Grounding Concept	A controlled grounding concept shall be used so that grounding will not adversely impact the functional testing/operation of the UUT. This concept shall provide for a single point ATE signal ground as well as a single point safety ground IAW the guidelines of MIL-HDBK-419.	TBD

GRA Ref.	Name	Text/Description	Verification
SRD_1.5.3.2	Shock Protection	The grounding scheme for the Tester shall provide protection of personnel and equipment from electrical shock hazards during normal operations and under ground fault conditions.	Analysis
SRD_1.5.3.3	Ground Loop Prevention	The Tester shall be designed to eliminate ground loops and minimize common ground impedance which can degrade system noise margins.	Analysis
SRD_1.5.3.4	External Safety Ground	The Intermediate Tester shall provide an external safety ground stud or bar.	Demonstration
SRD_1.5.4	Graphical Display		TBD
SRD_1.5.4.1	Graphical Display Size	The display shall be adequately sized to clearly show at a minimum all applicable information as defined in specification paragraph 3.1.2.2 (e.g., status, menus, test data, fault codes, messages, etc.) in a manner that is easily readable by the user. The display shall meet the requirements of MIL-STD-1472, Section 5.2.	TBD
SRD_1.5.4.2	Graphical Display Readability	The display shall be readable in all lighting conditions, including full daylight ambient conditions; direct sunlight; and night time IAW MIL-STD-1472G, Section 5.2. The brightness setting shall not allow the display to "black-out".	Demonstration
SRD_1.5.4.3	Graphical User Interface	The Intermediate Tester graphical user interface functions shall enable the operator to interact with the test system, to include but not be limited to, enabling TPS diagnostic advancement and/or referral to alternate test instructions, explanations and/or troubleshooting trees; authentication; test instruction initiation and/or termination; and test result save commands as necessary to achieve UUT test requirements.	Demonstration
SRD_1.5.4.4	User Navigation	The Tester shall provide controls to allow the operator to move between pages of display information, between selection options, and selection of a menu item/option.	Demonstration
SRD_1.5.4.5	Handheld	(U) Common Handheld Tester Display Size shall depend on final tester dimensions. Screen should follow current military standards with sunlight readable graphic display with resolution at least 128 X 64 pixels or better, and different screen background. The handheld tester display will be at a minimum 1.5 inches by 2.5 inches in Viewable Area.	Demonstration
SRD_1.6	Modular Open System Architecture Design		
SRD_1.6.1	Modular Design	b. Modular, Open Design – The contractor shall develop an architecture that is layered and modular and uses standards-based hardware interfaces, operating systems, and middleware that all utilize either non-proprietary or non-vendor-unique key module or component interfaces. The contractor's design approach shall be applied to all subsystems and components.	Analysis

GRA Ref.	Name	Text/Description	Verification
SRD_1.6.1.1	Module Cohesion	ii. Module Cohesion – The contractor’s design shall result in modules that are characterized by the singular assignment of identifiable and discrete functionality (high cohesion). The purpose is to ensure that any changes to system behavioral requirements can be accomplished by changing a minimum number of modules within the system.	Analysis
SRD_1.6.1.2	Module Coupling	i. Module Coupling – The contractor’s design approach shall result in modules that have minimal dependencies on other modules (loose coupling), as evidenced by simple, well-defined interfaces and by the absence of implicit data sharing. The purpose is to ensure that any changes to one module will not necessitate extensive changes to other modules, and hence facilitate module replacement and system enhancement.	Analysis
SRD_1.6.2	Open Architecture	a. Open Architecture – The contractor shall develop and maintain an architecture that incorporates appropriate considerations for re-configurability, portability, maintainability, technology insertion, vendor independence, reusability, scalability, interoperability, upgradeability, and long-term supportability.	Analysis
SRD_1.6.3	Requirements Accountability	c. System Requirements Accountability – The contractor shall ensure that all system requirements are accounted for through a demonstrated ability to trace each requirement to one or more modules that consist of components that are self-contained elements with well defined, open and published interfaces implemented using open standards.	Analysis
SRD_1.6.4	Inter-component Dependencies	d. Inter-component Dependencies – The contractor’s design approach shall result in a layered system design, maximizing software independence from the hardware, thereby facilitating technology refresh. The design shall be optimized at the lowest component level to minimize inter-component dependencies. The layered design shall also isolate the application software layers from the infrastructure software (such as the operating system) to enhance portability and to facilitate technology refresh. The design shall be able to survive a change to the computing infrastructure with minimal or no changes required to the application logic. The interfaces between the layers shall be built to open standards or the technical data describing the interface shall be Unlimited Data Rights. The system architecture shall minimize intercomponent dependencies to allow components to be decoupled and reused, where appropriate, across various DoD or Service programs and platforms.	Analysis
SRD_1.7	System Internal Data Requirements*		

GRA Ref.	Name	Text/Description	Verification
SRD_1.9	Environmental, Safety, and Operational Health requirements		
SRD_1.9.1	Materials	All material, processes and parts for the Tester shall conform to the requirements of MIL-PRF-28800, paragraph 3.2 excluding paragraph 3.2.4	Analysis
SRD_1.9.2	Acoustic Noise	Acoustic noise requirements shall be as specified in MIL-PRF-28800F, Paragraph 3.8.12. The Tester System shall not generate acoustic noise in excess of 70 dBA sound pressure level (SPL) when measured at the operator and bystander positions, IAW ISO 7779.	Test
SRD_1.9.2.1	Moving Parts	All moving/mechanical parts shall be inaccessible to the operator during system operation to prevent inadvertent contact (e.g., cooling fan).	Inspection Analysis
SRD_1.9.3	Explosive Atmosphere	The Tester shall be designed to be Explosive Atmosphere Qualified per MIL-STD-810, Method 511.6, Procedure I. The Tester design shall have the ability to operate in a fuel vapor environment without igniting the environment.	Test
SRD_1.9.4	Hardware Safety Engineering	Additional Hardware Safety shall be developed IAW MIL-STD-882.	Analysis
SRD_1.10	Security and Privacy Requirements		
SRD_1.10.1	Anti-Tamper (Reserved - Pending CPI Assessment)		
SRD_1.10.1.1	Anti-Tamper System Design	The contractor shall design and implement AT features into system hardware, software, and firmware and provide an updated final version of the AT Plan IAW Anti-Tamper Plan, DI-MISC-80508B/T Final.	TBD
SRD_1.10.1.2	Anti-Tamper Verification Plan	The Contractor shall deliver the initial AT Verification Plan 120 days prior to the execution of formal AT Verification Testing IAW Anti-Tamper Plan, DI-MISC-80508B/T.	TBD
SRD_1.10.1.3	AT Verification Test Procedures	The Contractor shall deliver the final AT Verification Test Procedures 60 days prior to the execution of formal witnessed AT Verification Testing IAW Anti-Tamper Verification Procedures, DI-MISC-80508B/T.	TBD
SRD_1.10.1.4	Anti-Tamper Verification Testing	The Contractor shall perform formal AT Verification Testing to verify the AT implementation architecture, design, and features protect system CPI.	TBD
SRD_1.10.1.5	Preliminary Anti-Tamper Architecture	The contractor shall develop a conceptual preliminary system level Anti-Tamper (AT) architecture and design to protect system CPI IAW Anti-Tamper Plan, DI-MISC-80508B/T Preliminary.	TBD

GRA Ref.	Name	Text/Description	Verification
SRD_1.10.1.6	Anti-Tamper Verification Testing Results	The Contractor shall document all AT Verification Testing results, including formal and informal verification activities, in an AT Verification Report, which will document all anomalies encountered during AT Verification Testing and provide detailed resolutions for each IAW Anti-Tamper Verification Report – DI-MISC-80508B/T.	TBD
SRD_1.10.1.7	Implementation Model Views of Anti-Tamper	The Contractor shall document the Anti-Tamper view of the Implementation Model as mapped to the TOSA Domains, hardware components and assemblies, software CSCIs, and firmware CSCIs IAW the Implementation Model Technical Material.	TBD
SRD_1.10.1.8	Protection of System Critical Program Information (CPI)	The Contractor shall protect Tester Critical Program Implementation (CPI) IAW DoDI 5200.39, “DoD AT Technical Implementation Guidance (TIG) (U),” (Secret), Defense Acquisition Guidebook Ch. 9 (latest version), DoDI 5229.27, DoDM S-5229.27 (Secret), and the latest version of the “DoD AT Desk Reference.”	TBD
SRD_1.10.2	Cyber Security	a. The Tester shall maintain a mission effective capability in the face of a full spectrum of cyber threats. b. If commercial operating systems and/or software is used, then applicable Defense Information Systems Agency (DISA) Security Technical Implementation Guide (STIG) shall be implemented. Non-commercial operating systems will be subject to generic cybersecurity STIGs and expectations. c. The requirements listed below shall be implemented in the design in a manner to minimize the impact on operational and maintenance procedures in the flight line environment.	TBD
SRD_1.10.2.1	User Auditing	If the system utilizes an operating system, it shall incorporate an electronic auditing trail that unambiguously identifies and records the successful and unsuccessful attempts to access/modify the system and its resources to include: authentication, account/group management, system configuration change, software change, audit change.	Test
SRD_1.10.2.2	System Recovery	The Tester shall be capable of being recovered to a known clean/good state.	Test
SRD_1.10.2.3	Malware Alerting	The Tester shall notify the user of any detected malware or unauthorized changes and not allow activation of the test connections if malware is detected during system start up.	Test
SRD_1.10.2.5	Least Functionality	The Tester shall only include necessary and functional hardware and software to accomplish the specific requirements of the system. This includes the removal of all unnecessary software and services of an operating system and the exclusion of unnecessary data ports or interfaces.	Test
SRD_1.10.2.6	System Administration	The Tester shall deny software configuration changes for all but administrator user accounts. The operating system, test executive, and TPS software shall only be capable of being loaded by those users who have administrator privileges. The Tester shall permit only authorized software code to execute on the system.	Test
SRD_1.10.2.7	Configuration Auditing	The Tester shall perform a configuration audit during system start to determine if changes have been made.	Test

GRA Ref.	Name	Text/Description	Verification
SRD_1.10.2.8	Data at Rest Encryption	The Tester shall encrypt all data at rest on the nonvolatile storage media used to boot the Tester, store the operating system, store the test executive and store the Test Program Set Software.	Test
SRD_1.10.2.9	Software Non-Repudiation	All Tester Software, TPSs, and firmware created, delivered, and installed shall utilize non-repudiation techniques to guarantee the integrity and origin of the System Software, TPSs and firmware throughout the entire System Software, TPSs and firmware lifecycle. (Examples: software hashing, digital signatures)	Test
SRD_1.10.2.10	Role based access control	If the system utilizes an operating system, it shall incorporate role-based access control mechanisms.	Test
SRD_1.10.2.11	Minimized Privileges	User accounts will have the minimal amount of privileges required for that account to perform its function.	Test
SRD_1.10.2.12	Supportability	No unsupported software such as old versions of operating systems (i.e. Windows XP) will be used on the Tester.	Test
SRD_1.10.2.13	Information Input Validation	All data inputted by the user should be validated to ensure it conforms with expected format and content.	Test
SRD_1.11	System Environment Requirements		
SRD_1.11.1	Altitude	The Tester shall conform to the specified performance and accuracy requirements when operated at an altitude 15,000 feet (4,600 meters) or less IAW MIL-STD-810G Method 500.6 Procedure II (operating configuration), and after return from an altitude of 40,000 feet, IAW MIL-STD-810G, Method 500.6, Procedure III (storage/transit configuration).	Test
SRD_1.11.2	System Management	The Tester shall be designed to be managed IAW AFMCI21-103 Reliability-Centered Maintenance (RCM) Programs or approved method of analysis designated by engineering or other driving guidance.	Analysis
SRD_1.11.6	Ruggedized Handheld CAT-F Tester	(U) The developed CAT-F will be designed to integrate survivability fundamentals to function reliably in harsh environments such as extremes of temperature, shock and vibration, humidity, or radiation. All CAT-F O-Level/flightline components (Handheld Tester, Preload Kit, Cables...) will meet MIL-PRF-28800G requirements for tailored Class I. The fault isolation kit will meet MIL-PRF-28800G requirements for a tailored Class II. Verified by digital twin representation and/or tester simulation in required operational conditions.	TBD
SRD_1.11.6.1	Fault Isolation Class II Tailoring	Splash Proof - Add language Need to add dust proof to I-level?	Test
SRD_1.11.6.2	Handheld Class I Tailoring	None currently defined	Test

GRA Ref.	Name	Text/Description	Verification
SRD_1.12	Computer Resource Requirements	Define and discuss software data rights (ensure can transition to Organic support-normally deemed Core).Does the program intend to utilize existing AF IT Systems to manage and sustain the system, sub-system, commodity, etc.	TBD
SRD_1.12.1	Computer software requirements		
SRD_1.12.1.3	Operating System Architecture	Operating System Software shall be designed with middle-ware to interface TPSs and the Test Executive Software with the operating system such that updates to operating system do not affect TPS qualification. The Tester operating system shall be supportable and/or upgradable over the service life of the tester without modifying TPSs.	Test
SRD_1.12.1.4	Test Executive Software Architecture	Test Executive Software shall be designed with middle-ware to interface TPSs with the operating system such that updates to operating system do not affect TPS qualification.	Test
SRD_1.12.1.5	Test Program Set Software Architecture	The Tester software environment shall allow TPSs to be instrument independent (hardware abstraction). Each TPS shall be a distinct configuration item, separate from the operating system and test executive, and not dependent upon other TPSs. TPSs shall be developed utilizing a contractor developed, Government approved Tester TPS style guide.	Test
SRD_1.12.2	Computer hardware requirements (Reserved)		
SRD_1.12.3	Computer communication requirements (Reserved)		
SRD_1.12.4	Computer hardware resource utilization requirements	Define % of utilization of the CPU within the Tester in worst case utilizations and worst case peak operations. All Computer Processing Units (CPUs) in the Tester shall maintain 50% excess capacity during worst case continuous utilization with a 50% excess capacity at worst case peak operations to ensure reliable operations and allow for growth.	Test
SRD_1.12.4.1	Memory Usage	The Tester shall be designed to utilize less than 50% of the system's volatile memory.	Test
SRD_1.12.4.2	Memory Capacity	The Tester shall have non-volatile memory large enough to store the operating system, TPS software, test results, and any supporting software with at least 50% open capacity for future growth.	Test
SRD_1.12.4.2.1	Electronic Storage	(U) Tester will be required to create and store testing results. The CAT-F shall create log files for all testing. The CAT-F must have ability to export test report log files to another computing device via direct connect. Tester will have minimum of 256 GB storage, able to store 7 days of data, and export log files. Greater than 256 GB storage and able to store 30+ days of data, and export log files is preferred.	Test

GRA Ref.	Name	Text/Description	Verification
SRD_1.13	System Quality Factors		
SRD_1.13.1	Reliability		
SRD_1.13.1.1	Tester Calibration	(U) The CAT-F will be designed to reduce downtime by eliminating the need for calibration or maximizing the time period between calibrations. Air Force Metrology and Calibration (AFMETCAL) initial certification calibration interval shall be longer than 12 months, but no calibration is preferred. If needed, calibration should be performed IAW T.O. 00-20-14.	Analysis
SRD_1.13.1.1.1	AFMETCAL CMRS	Contractor shall participate in Calibration Measurement Requirements Summary process IAW DID-QCIC-80278	TBD
SRD_1.13.1.2	Tester Reliability	(U) The CAT-F design will maintain an operational reliability in accordance with Military Standard MIL-PRF-28800F. Equipment shall be such that under normal use and operation the equipment does not fail within 1500 hours of operation with a statistical certainty of 95%. Reliability tests shall be performed by evaluation of manufacturer's data to establish that reliability falls within the parameters outlined.	Analysis
SRD_1.13.1.3	Mean Time Between Failure	The Tester operational Mean Time Between Failure (MTBF) shall be at least 1500 hours, with 95% confidence. MTBF is defined as the mean average time between failures. Express MTBF as: $MTBF = (\text{Operating Hours}) / (\text{Number of failures})$ A failure is any observed or indicated condition which requires the system to undergo unscheduled maintenance. A failure shall be counted for each separate root cause. Failures include software root causes, but exclude conditions induced by operator error. Failures may occur during both mission and non-mission time. Failures and indicated failures reported during inspection, or discovered by maintenance or calibration will be included in this metric.	Analysis
SRD_1.13.2	Maintainability		
SRD_1.13.2.1	Maintenance Concept	The Tester shall be designed for 2-level (organizational, intermediate, and depot) maintenance. Organizational level maintenance shall include replacement of minor externally accessible hardware. The contractor shall utilize respective MIL-HDBK 454 guidelines where applicable.	Analysis
SRD_1.13.2.4	Service Life	The Tester's service life shall be a minimum of 20 years with corrective maintenance. Service life begins at initial fielding.	Analysis
SRD_1.13.2.5	Storage Life	The Tester shall have a storage life of 5 years without maintenance.	Analysis
SRD_1.13.2.6	Accessibility	The Tester shall be designed to be accessible. MIL-HDBK-454 Guideline 36 shall be used as a guide.	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.13.2.7	Tester Repairability	(U) Tester should be modular design to allow unit repairability to maximum extent possible. Cables can be repaired at the field level to the maximum extent possible using current AF standard tools without vendor proprietary tooling. Tester components will be 80% repairable or replaceable at the unit level. All TPS cables will be equal or greater than 80% repairable and the cable connector assemblies must be fully repairable at base level. The design of the equipment shall permit isolation of faults and repair down to the component item or lowest replaceable unit with the maintenance provisions furnished. The Tester cables and interface adaptors shall be field-repairable. IAW T.O. numbers 1-1A-14 and 00-25-234. The Tester shall be capable of testing repaired cables post field actions.	Analysis
SRD_1.13.2.8	Scheduled Maintenance	The Tester shall be designed to minimize scheduled maintenance activities.	Analysis
SRD_1.13.2.9	Tester Power-On and Power Cycle Tracking	The Tester shall track and make note of the time the Tester was active (power-on), as well as its power on and off cycles.	Demonstration
SRD_1.14	Design and Construction Constraints		
SRD_1.14.1	Color	Exterior color shall be approved by the government	Analysis
SRD_1.14.2	Bonding	The Tester shall meet a minimum of System Bonding milliohms bonding throughout the electrical and grounding plan design, per tailored MIL-STD-464G system grounding and bonding requirements.	Inspection
SRD_1.14.3	Markings	All external devices which require an operational or maintenance interface shall be marked IAW MIL-STD-130. All markings shall be IAW DI-MGMT-81803A, DI-MGMT-81804A NOT 1, DI-MGMT-81858 NOT 1	Inspection
SRD_1.14.4	Enclosures	Enclosures for the Tester and cables/accessories shall be designed IAW MIL-PRF-28800G. Accessory enclosures shall prevent cable bends beyond manufacturer specification in storage.	Inspection
SRD_1.14.4.1	TPS Kit Weight	(U) The weight of each TPS kit shall not exceed 79 pounds per case inclusive of all cables, connectors, adapters, etc. that are contained within. The preferred Kit weight is less than 44 pounds using a dispatchable case with handles and wheels.	Inspection
SRD_1.14.5	Workmanship	MIL-HDBK-454 Guideline 9 shall be used as a guide to workmanship.	Analysis
SRD_1.14.6	Cable Design	Cables shall be designed to tolerate 90-degree flexing imposed by the weight of themselves during use, over their service life without damage to the internal wiring, cable exterior, connectors, or connector/cable interface IAW MIL-PRF-32070A. The cable branching limit imposed by MIL-PRF-32070A, Paragraph 3.4.4.2, is waived. The number of cable branches and connectors shall be minimized while optimizing size, weight, sustainment considerations and TPS run-time.	Analysis

GRA Ref.	Name	Text/Description	Verification
SRD_1.14.6.1	Wire Shields	Except for coaxial cables, wire shields, shall not be used as a return nor shall they be used to conduct power.	Inspection
SRD_1.14.6.2	Shield Bonding	Overall cable shields shall be terminated at a connector backshell that provides peripheral bonding of the shield.	Inspection
SRD_1.14.6.3	Protective Caps	Cables shall have a protective end-cap permanently attached by a lanyard and shall be ruggedized.	Inspection
SRD_1.14.6.4	Connector Finish	Connectors used to terminate shields shall have a conductive finish.	Inspection
SRD_1.14.6.5	Connector Keying	Connectors shall be keyed to prevent inadvertent application or use unless prevented by UUT.	Inspection
SRD_1.14.6.6	Shield Insulation	A layer of insulation shall cover all wire shields.	Inspection
SRD_1.14.6.7	Connector Selection	Connectors shall be selected using MIL-DTL-38999 and shall utilize environmentally sealed connectors with crimp style pins. Connectors that are not MIL-DTL-38999 compliant shall require government approval.	Inspection
SRD_1.14.6.8	Connector Population	To provide structural integrity when seating connectors, cable connectors shall be completely filled with pins and sockets.	Inspection
SRD_1.14.6.9	Connector Positioning	Cable connectors shall be positioned to mitigate strain and to facilitate ease of installation and/or removal without incurring ancillary accommodations (e.g., disconnection or removal of adjacent cables).	Inspection
SRD_1.14.6.10	Safety Ground Conductor	All power cables shall have a safety ground conductor IAW MIL-STD-464 and MIL-HDBK-419A.	Inspection
SRD_1.14.6.11	Cable Crosstalk Mitigation	Cables shall be designed to prevent crosstalk.	Inspection
SRD_1.14.6.12	I-LVL & O-LVL Cable Consolidation	(U) Backshop (I-LVL) and Flightline (O-LVL) Cables should incorporate standard cables for the same or equivalent UUT. I-LVL & O-LVL cable kits will be interchangeable and be a common configuration. If possible, the CAT-F tester would also be able to utilize legacy cables.	Inspection
SRD_1.14.7	EMI Emissions	The Tester shall meet the interface and associated verification requirements for the control of the Electromagnetic Interference (EMI) emission and susceptibility characteristics of electronic, electrical, and electromechanical equipment and subsystems per MIL-STD-461G Table V for Air Force Ground Equipment	Test
SRD_1.14.8	Fungal Resistance	The Tester shall incorporate materials which are non-nutritive and resist fungus growth IAW MIL-STD-810 Section 5.1	Analysis
SRD_1.14.9	Corrosion Prevention	The Tester shall be constructed of parts and materials that are corrosion and deterioration resistant, or coated to resist corrosion and deterioration, IAW MIL-STD-1568.	Inspection
SRD_1.14.10	Identification Plate	The system end items shall be properly identified IAW MIL-STD-130.	Inspection
SRD_1.14.14	Foreign Object Damage Prevention*	All parts and components shall be securely attached to prevent loss and to prevent Foreign Object Damage (FOD) to the associated Aircraft as well as the Tester or any other equipment.	Inspection

GRA Ref.	Name	Text/Description	Verification
SRD_1.14.15	Common Handheld Tester		
SRD_1.14.15.1	Tester Dimensions (Handheld)	(U) The CAT-F handheld Tester Dimensions shall accommodate required manual controls and readable screen. The total volume should be no more than 240 cubic inches (example L:12, H:5, and W:4 inches). Design shall be easily held in one hand by the user during operation.	Inspection
SRD_1.14.15.2	Common Handheld Tester Weight	(U) The CAT-F Handheld Tester shall be built with a light-weight handheld design and include a strap/strain relief (lanyard) which allows the handheld tester to be held or hung during extended use. Handheld tester shall not exceed more than 6 pounds with the batteries installed. The design must include strap/strain relief. The desired weight of the handheld tester is less than 3.5 pounds including batteries. The Handheld tester will be weighed to ensure that it is within specifications.	Inspection
SRD_1.14.15.3	Common Handheld Tester Display Size	(U) Common Handheld Tester Display Size shall depend on final tester dimensions. Screen should follow current military standards with sunlight readable graphic display with resolution at least 128 X 64 pixels or better, and different screen background. The handheld tester display will be at a minimum 1.5 inches by 2.5 inches in Viewable Area and display output shall be distinguishable to operator in all required environments.	Inspection
SRD_1.14.16	Test/Tester Set-up	(U) The CAT-F will be designed to reduce the number of connect / disconnect test steps for a single piece of armament equipment and allow for testing of multiple pieces of equipment simultaneously. The CAT-F tester set-up time when practical should not be more than 50% of testing time.	Demonstration
SRD_1.15	Personnel Related Requirements	Needs to be tailored for specific program.	TBD
SRD_1.15.1	Human Factors		
SRD_1.15.1.1	Transportability	The Transportability of the tester shall conform to requirements specified in MIL-STD-1366.	TBD

GRA Ref.	Name	Text/Description	Verification
SRD_1.15.1.1.1	Weight	The contractor shall strive to minimize the size, weight and total number of transit cases of the overall system without adversely affecting performance. The Tester footprint for the associated Aircraft shall not exceed that of the current legacy tester(s) and cable sets being replaced by the Tester for testing armament. The weight of the Tester including tester power cables and self-test cables or of any one accessory container for the Tester shall not exceed a two-person lift to 36 inches as defined for the “male only” population per MIL-STD-1472G, Section 5.8.6.3.1 (T) with an objective of a two-person lift to 36 inches as defined for the “male and female” population per MIL-STD-1472G, Section 5.8.6.3.1 or less (O).	Inspection
SRD_1.15.1.1.2	Handles	The Tester shall be provided with recessed and/or foldable handles or other suitable means for grasping, handling, and carrying IAW MIL-STD-1472G.	Inspection
SRD_1.15.1.1.3	Edges and Corners	The Tester shall include provisions for smooth edges/corners, prevent accidental contact with high and low surface temperatures, and provide pinch point prevention, IAW MIL-STD-1472G.	Inspection
SRD_1.15.1.2	Operability with Chemical Warfare Gear	The Tester shall be capable of being operated and decontaminated by personnel in Mission Oriented Protective Posture (MOPP) Level 4 Chemical Warfare Defense Ensemble (CWDE) IAW MIL-STD-1472G, Section 4.10.	Demonstration
SRD_1.16	Operational Requirements	List any other requirements for your program not explicitly listed within the requirements given.	TBD
SRD_1.17	Packaging Requirements	Items shall be packaged in accordance with MIL-STD-2073/1E, Standard Practice for Military Packaging. Shipping and storage markings shall be in accordance with MIL-STD-129R, Standard Practice for Military Marking. Items shall take into consideration approved KPP/KSA were applicable.	Inspection
SRD_1.18	Precedence and Criticality of Requirements	Create a table of Technical Performance Measures (TPM) and the goals associated with the TPMS set for the tester. Make sure important characteristics, such as reliability, weight, power, environmental, fault coverages, and other pertinent information are listed in the table. Order of preference is typically as follows: (1) Military; (2) Aerospace Materials Specification (AMS); (3) Federal; and (4) other acceptable industry documents.	TBD
SRD_1.18.1	Technical Performance Measures	The Contractor shall develop and deliver Technical Performance Measures for the system with event based schedules to indicate progress of TPMs.	TBD
SRD_1.19	Networking		
SRD_1.19.1	Network and Wireless Capable	The Intermediate Tester shall have provisions to communicate over networks and VPNs, IAW DODI 8420, TCPIP, VPN, and WIFI standards.	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.19.2	Networking Capability to Interface with FLoF	The Intermediate Tester shall interface with the Flight Line of Future (FLoF), and must be designed with the capability to communicate through commercial and critical interfaces. The Tester should be designed IAW RFC-0791, RFC-0793, IEEE 802.11xx (wifi), NIST/FIPS, DoDI 8100.02, AFMAN17-1301 (para 4.9), and AFMAN17-1302-O .	Demonstration
SRD_1.20	Software		
SRD_1.20.1	Runtime Services	Runtime services present within the Tester shall be designed IAW IEEE 1671 Annex D.	Analysis
SRD_1.20.2	Instrument Drivers	Drivers for instruments and software present within the Tester shall be designed IAW IVI-3.1 Driver Architecture Specification, IVI-3.2 Inherent Capabilities Specification, IVI-3.3 Standard Cross-Class Capabilities Specification, IVI-3.4 API Style Guide, IVI-3.14 Primary Interop Assembly Specification, as well as VPP-3.1 Instrument Driver Architecture and Design Specification, VPP-3.2 Instrument Driver Functional Body Specification, VPP-3.3 Instrument Driver Interactive Developer Interface Specification, and VPP-3.4 Instrument Driver Programmatic Developer Interface Specification.	Demonstration
SRD_1.20.3	ATE System Framework	Interfaces present within the system software shall be designed IAW VPP-2 System Frameworks Specification, IEEE 1671 ATMLI IVI-3.5 Configuration Server Specification, IVI-3.6 COM Session Factory Specification, IVI-3.9 C Shared Components Specification, IVI-3.10 Measurement and Stimulus Subsystems (MSS) Specification, IVI-3.12 Floating Point Services, IVI-3.15 IviLxiSync Specification.	Demonstration
SRD_1.20.4	Software Development Plan (SDP)	The Contractor shall develop and deliver the Software Development Plan for the Tester. The contractor shall provide a Software Requirements Specification (SRS), Software Design Document (SDD), Software Version Description (SVD), Software Test Plan (STP), Software User Manual (SUM), Software Implementation Document (SID), and any other appropriate documents IAW ATS processes.	Inspection
SRD_1.20.5	Instrument Communication Manager	Software present within the Tester shall be designed IAW VPP 4.3 Virtual Instrument Software Architecture.	Demonstration
SRD_1.21	Resource Management Services	All Tester signals shall be defined and described using IEEE 1641 Standard for Signal and Test Definition.	Inspection
SRD_1.22	Systems Engineering		
SRD_1.22.1	Cybersecurity Systems Engineering	Efforts to develop Cybersecurity principles via Systems Engineering shall be done IAW ACAR CSF Subcategories, CROWS CS Guidebook, Cyber Policy Model, and ATS Implementation.	Analysis
SRD_1.22.2	Systems Engineering and ILS Engineering	Contractor shall provide digital design data to a designated government Integrated Development Environment or provide access to a contractor Integrated Development Environment	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.22.3	Human Systems Integration (HSI) Engineering	Human Systems Integration shall be done IAW SAE 6906, MIL-STD-46855, MIL-PRF-38169, MIL-HDBK-759, and MIL-STD-1472.	Analysis
SRD_1.23	Program Management		
SRD_1.23.1	Integration of Data	Contractor shall provide digital design data to a designated government Integrated Development Environment or provide access to a contractor Integrated Development Environment	Demonstration
SRD_1.23.2	Risk Management Plan	The Contractor shall implement a Risk Management Plan for the Tester with documented risks and opportunities available to the Government for review and discussion from within the Implementation Model IAW DoDI 8510, and DI-MGMT-81808.	Demonstration
SRD_1.23.3	ILS Program Management	The management of Integrated Logistics Support (ILS) shall be conducted IAW DI-ILSS-80095.	Demonstration
SRD_1.23.4	Integrated Master Plan	The Integrated Master Plan for the Tester shall be developed IAW MIL-STD-881F.	Demonstration
SRD_1.23.5	Other Program Management	Other Program Management efforts shall refer to DI-MGMT-81797.	Demonstration
SRD_1.23.6	Integrated Master Schedule	The Integrated Master Schedule for the Tester shall be developed IAW DI-IPSC-82297.	Demonstration
SRD_1.23.7	Software Program Management	Relevant Software used and associated with the Tester shall be managed IAW IEEE-STD-12207, and DI-MGMT-81797.	Demonstration
SRD_1.23.8	Cost, Schedule, & Performance	Cost, Schedule, and Performance efforts shall be developed IAW DoDI 5000.02, DI-MGMT-80368A, MIL-STD-881F, DI-FNCL-80331A, and DI-MGMT-81991.	Demonstration
SRD_1.24	System Test & Evaluation		
SRD_1.24.1	Mock-ups/SILs	Mock-up/ System/Software Integration Laboratories (SILs) used for developmental test shall be IAW DI-SESS-81770 System/Software Integration Laboratory (SIL) Development and Management Plan, and MIL-STD-881, Appendix K.3.4.4 Work Breakdown Structures for Defense Materiel Items.	Inspection
SRD_1.24.2	Cybersecurity Test & Evaluation	The contractor will provide resources necessary to support independent cybersecurity test and evaluation events to include penetration test events and cybersecurity acceptance testing. Additionally, the contractor will provide results of developmental cybersecurity testing as part of the incremental development of the system. Cybersecurity Test and Evaluation of the systems will include SCAP Compliance Scanning of applicable operating systems and software, ACAS scanning of systems and static code analysis of all software developed by the contractor.	Inspection

GRA Ref.	Name	Text/Description	Verification
SRD_1.24.3	Developmental and Operational Test & Evaluation	Developmental and Operational Test & Evaluation procedures shall be developed IAW DoDI 5000.89_DAFI99-103 Capabilities-Based Test and Evaluation, MIL-STD-881, Appendix K.3.4.1 and K.3.4.2 Work Breakdown Structures for Defense Materiel Items, and DI-NDTI-81284 Test and Evaluation Program Plan.	Inspection
SRD_1.25	Data	Deliverables	
SRD_1.25.1	TPS Technical Data Package (TDP)	Technical Data Package shall be developed, produce, maintain, and delivered that accurately represent the approved, tested, and accepted configuration of the defined delivered item(s), in addition depict the final product baseline. Standards shall be IAW MIL-STD-3100B, TO 00-5-1, MIL-STD-38784 Rev B, and TO 00-5-3. The applicable manual(s) requirements shall contain specifications outlined in the approved AF Technical Manual Contract Requirement (TMCR) and shall prepare AFTO Forms 43 and 186. (i.e. TO's, Job Guides, IPB's, Manuals, etc.) If applicable, shall establish a Technical Order Library.	Inspection
SRD_1.25.2	Acquisition Sustainment Data Package (ASDP)	The Acquisition Sustainment Data Package shall be developed IAW DI-DRPR-81961C.	Inspection
SRD_1.25.2.1	Data Rights	Define the Data Rights your program expects from the contractor for documentation and other information related to the Tester. The Data Rights herein shall adhere to the IP Strategy Template, Data Rights and IP, and AFMCI 63-101 (checklist).	TBD
SRD_1.25.2.2	Data Repository	The GRA shall act as the Data Repository for the CAT-F program.	Analysis
SRD_1.25.2.3	Data Deliverables		
SRD_1.25.2.3.1	Hardware Bill of Material (BOM)	A comprehensive Bill of Material (BOM) shall be developed, comprised of all relevant components and accessories for the hardware used within the Tester and applicable standards outlined in DI-PSSS-81656 and IPS Elements, para 4.4, and MIL-PRF-49506.	Inspection
SRD_1.27.4.3.2	Software Bill of Material (BOM)	A comprehensive Bill of Material (BOM) shall be developed IAW DI-SESS-82433 and reference PSM para 7.1 & 12.1.2.2.1 for the software used within and to support the Tester.	Inspection
SRD_1.25.2.4	Interface Control Document (ICD)	The Interface Control Document (ICD) for the Tester shall be developed IAW MIL-HDBK-2222, DI-SESS-81248, and ATML Standards.	Inspection
SRD_1.25.2.5	Development Environment Documentation	The Development Environment Documentation for the Tester shall be developed IAW ATML Standards.	Inspection

GRA Ref.	Name	Text/Description	Verification
SRD_1.25.2.6	System/Software Integration Laboratories (SIL) Documentation	The System/Software Integration Laboratories (SIL) Documentation for the Tester shall be developed IAW ATML Standards.	Inspection
SRD_1.26	Processes		
SRD_1.26.1	Engineering Data Review		
SRD_1.26.1.1	SERDs	Contractor shall participate in Support Equipment Recommendation Data process IAW MIL-HDBK-2097	TBD
SRD_1.27	Methods		
SRD_1.27.1	MBSE	The Contractor shall employ a MBSE approach across the spectrum of systems engineering activities (e.g., requirements flow down, specifications development, System Engineering Master Plan development, risk management). This includes the use of the Object-Oriented Systems Engineering Method (OOSEM), the Systems Modeling Language (SysML), SysML-compatible MBSE tools, and utilizing the Model-Based Systems Architecture Process (MBSAP) methodology (i.e. capabilities, operational, logical, functional, physical views, system build, integration, test, and system prototypes).	Analysis
SRD_1.27.2	Style Guide for Modeling (Reserved-Pending Style Guide Development)		Contractor shall adhere to ATS Style Guide for MBSE
SRD_1.28	Tools		
SRD_1.28.1	MEARS		
SRD_1.28.2	Solid Works from Models	All 3D Models for the Tester and its associated components shall be compatible with SolidWorks modeling software.	TBD
SRD_1.28.3	Cameo SysML Modeling Tool	All System Modelling for the Tester and its associated components shall be compatible with Cameo/Catia Magic System of Systems SysML modelling software.	Demonstration
SRD_1.28.4	System Modeling Language (SysML) Standard	All System Models prepared for the Tester should adhere to SysML standards.	Demonstration
SRD_1.29	ILS Systems Engineering		
SRD_1.29.1	Supply Support	Supply Support shall be developed to address all management actions (SCM), procedures, and techniques to determine requirements. i.e. how to acquire, repair, repair spares, spare parts, supplies, and catalog, initial provisioning etc. PSM Handbook, para A.4, AFI63-101/20-101.	Analysis

GRA Ref.	Name	Text/Description	Verification
SRD_1.29.2	Technical Data	Technical Data Package shall be developed, produce, maintain, and delivered that accurately represent the approved, tested, and accepted configuration of the defined delivered item(s), in addition depict the final product baseline. Standards shall be IAW MIL-STD-31000B, TO 00-5-1, MIL-STD-38784 Rev B, and TO 00-5-3. The applicable manual(s) requirements shall contain specifications outlined in the approved AF Technical Manual Contract Requirement (TMCR) and shall prepare AFTO Forms 43 and 186. (i.e. TO's, Job Guides, IPB's, Manuals, etc.) If applicable, shall establish a Technical Order Library. The TDP format will be clearly stated in the contract with an understanding of how lifecycle maintenance of the TDP will be performed (i.e. SGML, S1000D, 3D model, etc.).	Inspection
SRD_1.29.3	Design Interface	Design Interfaces shall list how you will ensure quantitative design characteristics (reliability, maintainability, availability, etc.) integrate with supportability elements (12 product support elements). Design interface reflect relationship of system to product support. AFI63-101/20-101 para5.4.21, DODI 5000.02, PSM Guidebook, and MIL-HDBK-502A	Analysis
SRD_1.29.4	Support Equipment	Support Equipment shall be developed to the maximum extent as possible, common and interoperable with other services and across agencies. (i.e. SERD, strategy, management, allowances etc.) AFI 63-101/20-101 para7.17.	Inspection
SRD_1.29.5	Manpower & Personnel	Manpower & Personnel shall be prescribe within program requirements, milestone progression, and needed services IAW MIL-HDBK-502 while utilizing current manpower capability at maximum extent as possible.	Analysis
SRD_1.29.6	Sustaining Engineering	Sustaining Engineering shall identify support to the end item across the life cycle of the system. (i.e. once it's fielded, how will we provide engineering support, identification, review, assessment, resolution of deficiencies, analysis, TO updates, etc.) AFI63-101/20-101, para7.3, PSM Guidebook, para A.3.	Analysis
SRD_1.29.7	Product Support Management	Product Support Management shall take into account how IPS will be managed across life cycle of the system (full/partial CLS, ICS, organic, modifications etc.).	Analysis
SRD_1.29.8	Facilities & Infrastructure	Facilities & Infrastructure shall take into consideration or leverage existing facilities & infrastructure, (i.e. hangers, AF owned industrial facilities, SIL, inclusive of O/I/D Levels, information technology (software). IAW AFMCI 21-109, PSM para A.11, Integrated Product Support (IPS) Guidebook page 285	Analysis
SRD_1.29.9	Training & Training Support	Training & Training Support shall be developed to the maximum extent as possible. (i.e. reduce cost, aligned with other key processes, decentralize locations, accommodate multiple MDS, optimize train-the trainer, virtual etc.) DI-ILSS-80872, PSM para 9 & 11.	Demonstration

GRA Ref.	Name	Text/Description	Verification
SRD_1.29.10	Maintenance Planning & Management	Maintenance Planning & Management shall be developed to define maintenance concepts, procedures, and requirements for the life of the system for both hardware and software. i.e. levels of repair, upkeep of task, schedule, sustain system, components, SE, , etc. PSM para A.5, DODI 5000.91, para 4.2, AFI63-101/20-101, para 7.13	Analysis
SRD_1.29.11	Packaging, Handling, Storage, & Transportation (PHST)	Packaging, Handling, Storage, & Transportation shall be developed IAW military or commercial standards. AFMCI24-201, AFI63-101/20-101 para5 .4.19 and MIL-STD-2073/1E, PSM para A.6.	Inspection

ATTACHMENT 2:

Common Armament Tester for Fighter (CAT-F)

F-16/F-15/A-10 ALTERNATE MISSION EQUIPMENT (AME)

Prepared by:
ATS Engineering Office
AFMC AFLCMC/WNA ATS Program Office
Robins AFB, Georgia
15 July 2024

The Common Armament Tester for Fighter (CAT-F) Alternate Mission Equipment (AME) list helps defines the pylons, launchers, or bomb racks used by the CAT-F fighter Aircraft.

CAT-F ALTERNATE MISSION EQUIPMENT	
F-16 ALTERNATE MISSION EQUIPMENT (O-Level)	
LAU-129 Launcher	TER-9A
Non-1760 Wing Weapons Pylon	Gun Control Unit (M61A1)
Non-1760 PIDS Wing Weapons Pylon	BRU-57/A with two BRU-46 Bomb Racks installed
1760 Wing Weapons Pylon	LAU-118
1760 PIDS Wing Weapons Pylon	MIL-STD-1760 Umbilical
Block 30 Centerline Pylon	370-gallon External Fuel Tank/Jettisonable Pylon
Non-1760 Centerline Pylon	LAU-131/LAU-68 Rocket Pod/Launcher Umbilical
1760 Centerline Pylon	Left Hardpoint Interface
MAU-12	Right Hardpoint Interface
F-16 ALTERNATE MISSION EQUIPMENT (I-Level)	
Under Wing Adapter	Non-1760 Centerline Pylon
ALE-50 Pylon	1760 Centerline Pylon
LAU-129 Launcher	MAU-12
Non-1760 Wing Weapons Pylon	TER-9A
Non-1760 PIDS Wing Weapons Pylon	Gun Control Unit (M61A1)
1760 Wing Weapons Pylon	BRU-57/A with two BRU-46 Bomb Racks installed
1760 PIDS Wing Weapons Pylon	BRU-46
Block 30 Centerline Pylon	LAU-118
F-15 ALTERNATE MISSION EQUIPMENT (O-Level)	
Pylon	1760 SUNW
Multiple Stores	Multipurpose Booster (MPB)
SUU-20	ALE-58A
BRU-46/47	SUU-60B/A Centerline Pylon
BRU-46/47 CFT	LAU-106A/A
BRU-61/A	LAU-128A/A, B/A, C/A, D/A
P-4A/AM AIS Pod	Missile Launcher Auxiliary Relay
IRST Pod	
F-15 ALTERNATE MISSION EQUIPMENT (I-Level)	
SUU-20	SUU-60B/A Centerline Pylon
BRU-46/47	SUU-59B/A Centerline Pylon
BRU-46/47 CFT	SUU-59C/A Inboard Pylon
BRU-61/A	SUU-73/A Centerline Pylon
MAU-12C/A, D/B	LAU-106A/A
1760 SUNW	LAU-128A/A, B/A, C/A, D/A
Multipurpose Booster (MPB)	

CAT-F ALTERNATE MISSION EQUIPMENT	
A-10 ALTERNATE MISSION EQUIPMENT (O-Level)	
Stations 1&11 Pylon (MAU-50 installed)	TER-9/A or Variant installed
Stations 2&10 Pylon (MAU-50 installed)	Dual Rail Adapter (DRA), LAU-105s and LAU-128s installed
Stations 3&9 Pylon (MAU-40 installed)	Dual Rail Adapter (DRA) LAU-105s and GMIU installed
Stations 4&8 Pylon (MAU-40 installed)	LAU-131
Station 5 Pylon (MAU-40 installed)	Chaff Flare
Station 6 Pylon (MAU-40 installed)	Gun Control Unit/Gun System
Station 7 Pylon (MAU-40 installed)	
A-10 ALTERNATE MISSION EQUIPMENT (I-Level)	
Stations 1&11 Pylon (MAU-50 installed)	Dual Rail Adapter (DRA) LAU-105s installed and LAU-128s installed
Stations 2&10 Pylon (MAU-50 installed)	Dual Rail Adapter (DRA) LAU-105s installed and GMIU installed
Stations 3&9 Pylon (MAU-40 installed)	LAU-131
Stations 4&8 Pylon (MAU-40 installed)	LAU-117
Station 5 Pylon (MAU-40 installed)	BRU-61
Station 6 Pylon (MAU-40 installed)	LAU-88
Station 7 Pylon (MAU-40 installed)	Pylon SCU-A/SCU-B
TER-9/A or Variant installed	

ATTACHMENT 3:
Common Armament Tester for Fighter (CAT-F)
ALTERNATE MISSION EQUIPMENT (AME)
MAINTENANCE LEVEL REQUIREMENTS

Prepared by:
ATS Engineering Office
AFMC AFLCMC/WNA ATS Program Office
Robins AFB, Georgia
15 July 2024

The United States Air Force CAT-F Alternate Mission Equipment (AME) Maintenance Level Requirements for the CAT-F program.

Common Armament Tester for Fighter (CAT-F)

CAT-F AME Maintenance Level Requirements	
F-16 AME Requirements	MX Level
TER-9/A TRIPLE EJECTOR RACK	
TER-9A (SWT)	O-Level
TER-9 (SST)	I-Level
TER BREECH AND ARMING LOOP TEST (500)	I-Level
LAU-117	
LAU-117 (V) 3/A LAUNCHER ELECTRONIC ASSEMBLY (LEA) (SWT)	O-Level
LAU-117 (V) 3/A FULL-UP (SST)	I-Level
LAU-117 (V) 3/A FULL-UP (500)	I-Level
LAU-117 (V) 3/A W/O LEA (SST)	I-Level
LAU-117 (V) 3/A W/O LEA (500)	I-Level
LAU-117 (V) 3/A LAUNCHER ELECTRONIC ASSEMBLY (LEA) (SST)	I-Level
LAU-117 (V) 3/A LAUNCHER ELECTRONIC ASSEMBLY (LEA) (500)	I-Level
LAU-118	
LAU-118 (SWT)	O-Level
LAU-118 FULL UP (SST)	I-Level
LAU-118 FULL UP (500)	I-Level
LAU-129	
LAU-129 (SWT)	O-Level
LAU-129 FULL UP (SST)	I-Level
LAU-129 FULL UP (500)	I-Level
LAU-129 FULL UP (SMS ARMADILLO)	I-Level
LAU-129 W/O AMRIU (SST)	I-Level
LAU-129 W/O AMRIU (500)	I-Level
LAU-129 W/O AMRIU, BIT (SST)	I-Level
LAU-129 W/O AMRIU, BIT (500)	I-Level
LAU-129 W/O AMRIU, BIT, PS (SST)	I-Level
LAU-129 W/O AMRIU, BIT, PS (500)	I-Level
LAU-129 W/O AMRIU, BIT, PS, ASU (SST)	I-Level
LAU-129 W/O AMRIU, BIT, PS, ASU (500)	I-Level
I-LEVEL LAU-129 W/O AMRIU, BIT, PS, ASU, PSU (SST)	I-Level
LAU-129 W/O AMRIU, BIT, PS, ASU, PSU (500)	I-Level
LAU-129 W/O AMRIU, BIT, PS, ASU, PSU, DETENT (SST)	I-Level
LAU-129 W/O AMRIU, BIT, PS, ASU, PSU, DETENT (500)	I-Level
LAU-129 DETENT FLY THRU CHECK (SST)	I-Level
LAU-129 DETENT FLY THRU CHECK (500)	I-Level
LAU-129 POWER SUPPLY (SST)	I-Level
LAU-129 POWER SUPPLY (500)	I-Level
LAU-129 AUDIO SWITCH UNIT (SST)	I-Level
LAU-129 AUDIO SWITCH UNIT (500)	I-Level
I-LEVEL LAU-129 POWER SWITCHING UNIT (SST)	I-Level
LAU-129 POWER SWITCHING UNIT (500)	I-Level
LAU-129 BUILT IN TEST (SST)	I-Level
LAU-129 BUILT IN TEST (500)	I-Level

Common Armament Tester for Fighter (CAT-F)

CAT-F AME Maintenance Level Requirements	
F-16 AME Requirements	MX Level
CENTERLINE PYLON WITH MAU-12 INSTALLED	
CENTERLINE PYLON WITH MAU-12 INSTALLED (SWT)	O-Level
BLK 50 1760 CENTERLINE PYLON FULL-UP (SST)	I-Level
BLK 50 1760 CENTERLINE PYLON FULL-UP (500)	I-Level
BLK 50 1760 CENTERLINE PYLON W/O ACRIU (SST)	I-Level
BLK 50 1760 CENTERLINE PYLON W/O ACRIU (500)	I-Level
BLK 50 1760 CENTERLINE PYLON W/O MAU12 (SST)	I-Level
BLK 50 1760 CENTERLINE PYLON W/O MAU12 (500)	I-Level
BLK 50 1760 CENTERLINE PYLON HARNESS (SST)	I-Level
BLK 50 1760 CENTERLINE PYLON HARNESS (500)	I-Level
UNDERWING ADAPTER (SST)	I-Level
UNDERWING ADAPTER (500)	I-Level
WING WEAPONS PYLON WITH MAU-12 INSTALLED	
WING WEAPONS PYLON WITH MAU-12 INSTALLED (SWT)	O-Level
1760 WING WEAPON PYLON FULL-UP (SST)	I-Level
1760 WING WEAPON PYLON FULL-UP (500)	I-Level
1760 WING WEAPON PYLON W/O ACRIU (SST)	I-Level
1760 WING WEAPON PYLON W/O ACRIU (500)	I-Level
1760 WING WEAPON PYLON W/O MAU-12 (SST)	I-Level
1760 WING WEAPON PYLON W/O MAU-12 (500)	I-Level
1760 WING WEAPON PYLON HARNESS (SST)	I-Level
1760 WING WEAPON PYLON HARNESS (500)	I-Level
MAU-12 STANDALONE (SST)	I-Level
MAU-12 STANDALONE (500)	I-Level
MIL-STD-1760 UMBILICAL	
MIL-STD-1760 UMBILICAL (SWT)	O-Level
BRU-57 WITH BRU-46 INSTALLED	
BRU-57 WITH BRU-46 INSTALLED (SWT)	O-Level
BRU-57/A CSSI 1 BOMB RACK-UAI (SST)	I-Level
BRU-57/A CSSI 2 BOMB RACK-UAI (SST)	I-Level
BRU-46/A BOMB RACK (SST)	I-Level
370 TANKS (FLETCHER AND NON-FLETCHER)	
370 TANKS (FLETCHER AND NON-FLETCHER) (SWT)	O-Level
M61A1 GUN SYSTEM	
M61A1 GUN SYSTEM (SWT)	O-Level
GUN CONTROL UNIT (500)	I-Level
LAU-131/LAU-68 ROCKET LAUNCHER UMBILICAL	
LAU-131/LAU-68 ROCKET LAUNCHER UMBILICAL (SWT)	O-Level
SUU-20	
SUU-20 (SWT)	O-Level
SUU-20 (REHOST FROM F-15 T-197 TESTER) (500)	I-Level
CLEAN STATION TESTING	
AIR TO GROUND STATIONS 3, 4, 6 AND 7 (SMS ARMADILLO)	O-Level
AIR TO AIR STATIONS 1, 2, 3A, 7A, 8 AND 9 (SMS ARMADILLO)	O-Level
CENTERLINE INTERFACE STATION 5 (SMS ARMADILLO)	O-Level
GCU GUN TEST (SMS ARMADILLO)	O-Level
LEFT AND RIGHT HARDPOINTS (5R, 5L) (SMS ARMADILLO)	O-Level

Common Armament Tester for Fighter (CAT-F)

CAT-F AME Maintenance Level Requirements	
F-15 AME Requirements	MX Level
JETTISON CHECKS W/O FUEL TANKS	
(WITH FAULT ISOLATION CAPABILITY OF ALL LISTED) (T-198 SMSTS MSIP) (C/D)	O-Level
PYLON JETT CHECK (T-199 ACPTS MSIP) (C/D)	O-Level
PYLON JETT (T-199 ACPTS) (E)	O-Level
PYLON JETT (T-199 ACPTS) (EX)	O-Level
PYLON JETT (T-198 SMSTS MSIP) (C/D)	O-Level
PYLON JETT (T-198 SMSTS) (E)	O-Level
PYLON JETT (T-198 SMSTS) (EX)	O-Level
MAU12 BOMB RACK (T-197 SMSSTS)	I-Level
MAU12 JETT CHECK (T-199 ACPTS MSIP) (C/D)	O-Level
MAU12 JETT CHECK (T-199 ACPTS) (E)	O-Level
PYLON SINGLE STORES (T-198 SMSTS MSIP) (C/D)	O-Level
PYLON SINGLE STORES (T-198 SMSTS) (E)	O-Level
PYLON SINGLE STORES	O-Level
JETTISON CHECKS W/ FUEL TANKS	
MAU12 SINGLE STORES (T-199 ACPTS MSIP) (C/D)	O-Level
(T-199 ACPTS) (E)	O-Level
JETTISON CHECKS	
PYLON JETT CHECK (T-199 ACPTS MSIP) (C/D)	O-Level
PYLON JETT (T-199 ACPTS) (E)	O-Level
PYLON JETT (T-199 ACPTS) (EX)	O-Level
PYLON JETT (T-198 SMSTS MSIP) (C/D)	O-Level
PYLON JETT (T-198 SMSTS) (E)	O-Level
PYLON JETT	O-Level
BRU-46/47 BREECH (CFT) (PYLON)	
BRU46/47 JETT CHECK (T-199 ACPTS) (E)	O-Level
BRU46/47 JETT CHECK (T-199 ACPTS) (EX)	O-Level
CFT SINGLE STORES (T-198 SMSTS) (E)	O-Level
CFT SINGLE STORES	O-Level
CFT AIR/GROUND TESTS (T-197 SMSSTS)	I-Level
(T-199 ACPTS) (E)	O-Level
CFT AIR/AIR TESTS (T-197 SMSSTS)	I-Level
(T-199 ACPTS) (E)	O-Level
CFT 1760 TESTS (T-197 SMSSTS)	I-Level
AIM-9M MISSILE SYSTEM	
(T-199 ACPTS) (E)	O-Level
CFT SPL WPN TESTS (T-197 SMSSTS)	I-Level
AIM-9 (T-199 ACPTS MSIP) (C/D)	O-Level
AIM-9 (T-199 ACPTS) (E)	O-Level
AIM-9 (T-199 ACPTS) (EX)	O-Level
AIM-9 (T-198 SMSTS MSIP) (C/D)	O-Level
AIM-9 (T-198 SMSTS) (E)	O-Level
AIM-9	O-Level
AIM-9X/X-2 MISSILE SYSTEM	
(T-199 ACPTS) (E)	O-Level
AIM-9 HEAD SLAVE (T-198 SMSTS MSIP) (C/D)	O-Level
AIM-9 HEAD SLAVE (T-198 SMSTS) (E)	O-Level
AIM-9 HEAD SLAVE	O-Level

Common Armament Tester for Fighter (CAT-F)

CAT-F AME Maintenance Level Requirements	
F-15 AME Requirements	MX Level
AIM-120 LAU-128 Missile System	
AIM-9X (T-199 ACPTS MSIP) (C/D)	O-Level
AIM-9X (T-199 ACPTS) (E)	O-Level
AIM-9X (T-199 ACPTS) (EX)	O-Level
AIM-9X (T-198 SMSTS MSIP) (C/D)	O-Level
AIM-9X (T-198 SMSTS) (E)	O-Level
AIM-9X	O-Level
AIM-120 LAU-106 Missile System	
AIM-120 at LAU-128 (T-199 ACPTS MSIP) (C/D)	O-Level
AIM-120 at LAU-128 (T-199 ACPTS) (E)	O-Level
AIM-120 at LAU-128 (T-199 ACPTS) (EX)	O-Level
AIM-120 at LAU-128 (T-198 SMSTS MSIP) (C/D)	O-Level
AIM-120 at LAU-128 (T-198 SMSTS) (E)	O-Level
AIM-120 at LAU-128	O-Level
LAU-128A/A Launcher	
AIM-120 at LAU-106 (T-199 ACPTS MSIP) (C/D)	O-Level
AIM-120 at LAU-106 (T-199 ACPTS) (E)	O-Level
AIM-120 at LAU-106 (T-199 ACPTS) (EX)	O-Level
AIM-120 at LAU-106 (T-198 SMSTS MSIP) (C/D)	O-Level
AIM-120 at LAU-106 (T-198 SMSTS) (E)	O-Level
AIM-120 at LAU-106	O-Level
LAU-106A/A Launcher (T-197 SMSSTS)	I-Level
BRU-46/47 Single Carriage (CFT)	
LAU-106 SV Check (T-199 ACPTS MSIP) (C/D)	O-Level
LAU-106 Jett Check (T-199 ACPTS) (E)	O-Level
LAU-106 Jett Check (T-199 ACPTS) (EX)	O-Level
BRU-46/47 Single Carriage (Pylon)	
BRU46/47 Single Stores (T-199 ACPTS) (E)	O-Level
BRU46/47 Single Stores (T-199 ACPTS) (EX)	O-Level
BRU-61 UAI System Check	
BRU46/47 Single Stores (T-199 ACPTS) (E)	O-Level
BRU46/47 Single Stores (T-199 ACPTS) (EX)	O-Level
BRU46/A Bomb Rack (T-197 SMSSTS)	I-Level
CBU 103, 104, 105, 107, 118 Check	
GBU-39 UAI (T-199 ACPTS) (E)	O-Level
GBU-39 UAI (T-199 ACPTS) (EX)	O-Level
GBU-39 UAI (T-198 SMSTS) (E)	O-Level
GBU-39 UAI	O-Level
GBU-31 System Check	
CBU 103, 104, 105, 107, 118 Check (T-199 ACPTS) (E)	O-Level
CBU 103, 104, 105, 107, 118 Check (T-199 ACPTS) (EX)	O-Level
CBU 103, 104, 105, 107, 118 Check (T-198 SMSTS) (E)	O-Level
CBU 103, 104, 105, 107, 118 Check	O-Level
GBU-31B UAI System Check	
GBU-31 (T-199 ACPTS) (E)	O-Level
GBU-31 (T-199 ACPTS) (EX)	O-Level
GBU-31 (T-198 SMSTS) (E)	O-Level
GBU-31	O-Level

Common Armament Tester for Fighter (CAT-F)

CAT-F AME Maintenance Level Requirements	
F-15 AME Requirements	MX Level
AIM-9X/X-2 Missile System	
(T-199 ACPTS) (E)	O-Level
AIM-9 Head Slave (T-198 SMSTS MSIP) (C/D)	O-Level
AIM-9 Head Slave (T-198 SMSTS) (E)	O-Level
AIM-9 Head Slave	O-Level
AIM-120 LAU-128 Missile System	
AIM-9X (T-199 ACPTS MSIP) (C/D)	O-Level
AIM-9X (T-199 ACPTS) (E)	O-Level
AIM-9X (T-199 ACPTS) (EX)	O-Level
AIM-9X (T-198 SMSTS MSIP) (C/D)	O-Level
AIM-9X (T-198 SMSTS) (E)	O-Level
AIM-9X	O-Level
AIM-120 LAU-106 Missile System	
AIM-120 at LAU-128 (T-199 ACPTS MSIP) (C/D)	O-Level
AIM-120 at LAU-128 (T-199 ACPTS) (E)	O-Level
AIM-120 at LAU-128 (T-199 ACPTS) (EX)	O-Level
AIM-120 at LAU-128 (T-198 SMSTS MSIP) (C/D)	O-Level
AIM-120 at LAU-128 (T-198 SMSTS) (E)	O-Level
AIM-120 at LAU-128	O-Level
LAU-128A/A Launcher	
AIM-120 at LAU-106 (T-199 ACPTS MSIP) (C/D)	O-Level
AIM-120 at LAU-106 (T-199 ACPTS) (E)	O-Level
AIM-120 at LAU-106 (T-199 ACPTS) (EX)	O-Level
AIM-120 at LAU-106 (T-198 SMSTS MSIP) (C/D)	O-Level
AIM-120 at LAU-106 (T-198 SMSTS) (E)	O-Level
AIM-120 at LAU-106	O-Level
LAU-106A/A Launcher (T-197 SMSSTS)	I-Level
BRU-46/47 Single Carriage (CFT)	
LAU-106 SV Check (T-199 ACPTS MSIP) (C/D)	O-Level
LAU-106 Jett Check (T-199 ACPTS) (E)	O-Level
LAU-106 Jett Check (T-199 ACPTS) (EX)	O-Level
BRU-46/47 Single Carriage (Pylon)	
BRU46/47 Single Stores (T-199 ACPTS) (E)	O-Level
BRU46/47 Single Stores (T-199 ACPTS) (EX)	O-Level
BRU-61 UAI System Check	
BRU46/47 Single Stores (T-199 ACPTS) (E)	O-Level
BRU46/47 Single Stores (T-199 ACPTS) (EX)	O-Level
BRU46/A Bomb Rack (T-197 SMSSTS)	I-Level
CBU 103, 104, 105, 107, 118 Check	
GBU-39 UAI (T-199 ACPTS) (E)	O-Level
GBU-39 UAI (T-199 ACPTS) (EX)	O-Level
GBU-39 UAI (T-198 SMSTS) (E)	O-Level
GBU-39 UAI	O-Level
GBU-31 System Check	
CBU 103, 104, 105, 107, 118 Check (T-199 ACPTS) (E)	O-Level
CBU 103, 104, 105, 107, 118 Check (T-199 ACPTS) (EX)	O-Level
CBU 103, 104, 105, 107, 118 Check (T-198 SMSTS) (E)	O-Level
CBU 103, 104, 105, 107, 118 Check	O-Level

Common Armament Tester for Fighter (CAT-F)

CAT-F AME Maintenance Level Requirements	
F-15 AME Requirements	MX Level
GBU-31B UAI System Check	
GBU-31 (T-199 ACPTS) (E)	O-Level
GBU-31 (T-199 ACPTS) (EX)	O-Level
GBU-31 (T-198 SMSTS) (E)	O-Level
GBU-31	O-Level
M61A1 Gun (Internal Gun)	
GBU 31 UAI (T-199 ACPTS) (E)	O-Level
GBU 31 UAI (T-199 ACPTS) (EX)	O-Level
GBU 31 UAI (T-198 SMSTS) (E)	O-Level
SUU-20 Dispenser	
20 MM Gun (T-199 ACPTS MSIP) (C/D)	O-Level
20 MM Gun (T-199 ACPTS) (E)	O-Level
20 MM Gun (T-199 ACPTS) (EX)	O-Level
20 MM Gun (T-198 SMSTS MSIP) (C/D)	O-Level
20 MM Gun (T-198 SMSTS) (E)	O-Level
20 MM Gun	O-Level
SUU20	
SUU20 (T-199 ACPTS MSIP) (C/D)	O-Level
SUU20 (T-199 ACPTS) (E)	O-Level
SUU20 (T-199 ACPTS) (EX)	O-Level
SUU20 (T-198 SMSTS MSIP) (C/D)	O-Level
SUU20 (T-198 SMSTS) (E)	O-Level
SUU20	O-Level
SUU20B/A Dispenser (T-197 SMSSTS)	I-Level
Weapons System Check (Pylon)(CFT)	
Unknown (T-199 ACPTS) (E)	O-Level
Unknown (T-199 ACPTS) (EX)	O-Level
Unknown (T-198 SMSTS) (E)	O-Level
GBU-38UAI/-54 UAI System Check	
GBU-38 UAI (T-199 ACPTS) (E)	O-Level
GBU-38 UAI (T-199 ACPTS) (EX)	O-Level
GBU-38 UAI (T-198 SMSTS) (E)	O-Level
GBU-38 UAI	O-Level
GBU-53 UAI (T-199 ACPTS) (E)	O-Level
GBU-53 UAI (T-199 ACPTS) (EX)	O-Level
GBU-53 UAI (T-198 SMSTS) (E)	O-Level
GBU-53 UAI	O-Level
AGM-158 UAI	
AGM-158 (T-199 ACPTS) (E)	O-Level
AGM-158 (T-199 ACPTS) (EX)	O-Level
AGM-158 (T-198 SMSTS) (E)	O-Level
AGM-158	O-Level

Common Armament Tester for Fighter (CAT-F)

CAT-F AME Maintenance Level Requirements	
F-15 AME Requirements	MX Level
Additional AME	
Pylon SV Check (T-199 ACPTS MSIP) (C/D)	O-Level
Multi-Purpose Booster (T-199 ACPTS) (E)	O-Level
Pylon SV Check (T-199 ACPTS) (EX)	O-Level
P4AX Pod (T-198 SMSTS MSIP) (C/D)	O-Level
P4AX Pod (T-198 SMSTS) (E)	O-Level
P4AX Pod	O-Level
ADU-552/A Adapter (T-197 SMSSTS)	I-Level
Pylon SV Check (T-199 ACPTS) (E)	O-Level
EAIS Pod (T-198 SMSTS MSIP) (C/D)	O-Level
EAIS Pod (T-198 SMSTS) (E)	O-Level
EAIS Pod	O-Level
ADU-552B/A Adapter (T-197 SMSSTS)	I-Level
B61-5 (System 1) (T-199 ACPTS) (E)	O-Level
ALE-58 (T-198 SMSTS MSIP) (C/D)	O-Level
Multi-Purpose Booster (T-198 SMSTS) (E)	O-Level
CFT Multi Store	O-Level
SUU-59B/A Inboard Pylon (T-197 SMSSTS)	I-Level
B61-12 (System 2) (T-199 ACPTS) (E)	O-Level
MER (T-198 SMSTS MSIP) (C/D)	O-Level
CFT Multi Store (T-198 SMSTS) (E)	O-Level
ESU Aircraft Check	O-Level
SUU-59C/A Inboard Pylon (T-197 SMSSTS)	I-Level
IRST (T-198 SMSTS MSIP) (C/D)	O-Level
ESUAircraftt Check (T-198 SMSTS) (E)	O-Level
MER	O-Level
SUU-59D/A Inboard Pylon (T-197 SMSSTS)	I-Level
MER (T-198 SMSTS) (E)	O-Level
IRST	O-Level
SUU-60B/A Centerline Pylon (T-197 SMSSTS)	I-Level
B61-5 (System 1) (T-198 SMSTS) (E)	O-Level
SUU-60B/A PASS-modified Centerline Pylon (T-197 SMSSTS)	I-Level
B61-12 (System 2) (T-198 SMSTS) (E)	O-Level
SUU-73/A Centerline Pylon (T-197 SMSSTS)	I-Level
SUU-73B/A Centerline Pylon (T-197 SMSSTS)	I-Level
SUU-90/A Outboard Pylon (T-197 SMSSTS)	I-Level
SUU-115/A IRST Centerline Pylon (T-197 SMSSTS)	I-Level
MX-12427/A Dual Rail Outboard Adapter (T-197 SMSSTS)	I-Level
Encoder/Decoder/Power Supply (T-197 SMSSTS)	I-Level
Jumper Cable W67484DA (1760) (T-197 SMSSTS)	I-Level
Jumper Cable W67484FA (1760) (T-197 SMSSTS)	I-Level
Jumper Cable W67485AA (1760) (T-197 SMSSTS)	I-Level
Jumper Cable W67485CA (1760) (T-197 SMSSTS)	I-Level
Jumper Cable W67485DA (1760) (T-197 SMSSTS)	I-Level
Jumper Cable W67485GA (1760) (T-197 SMSSTS)	I-Level
Jumper Cable W67505AA (1760) (T-197 SMSSTS)	I-Level
Jumper Cable W67484AA (1760) (T-197 SMSSTS)	I-Level
Jumper Cable W67484CA (1760) (T-197 SMSSTS)	I-Level
Jumper Cable W68279 (SUU/MER) (T-197 SMSSTS)	I-Level

Common Armament Tester for Fighter (CAT-F)

CAT-F AME Maintenance Level Requirements	
A-10 AME Requirements	MX Level
Stations 1&11 Pylon (with MAU-50 installed)	
Stations 1&11 Pylon (with MAU-50 installed) (SVT/DVD)	O-Level
Clean Station Clean Station 1 (PATS-70)	O-Level
Pylon (PATS-70)	I-Level
Stations 1&11 Pylon (SVT/DVD)	O-Level
Clean Station Clean Station 11 (PATS-70)	O-Level
MAU-50(PATS-70)	I-Level
Stations 2&10 Pylon (with MAU-50 installed)	
Stations 2&10 Pylon (with MAU-50 installed) (SVT/DVD)	O-Level
Clean Station Clean Station 2 (PATS-70)	O-Level
Pylon (PATS-70)	I-Level
Stations 2&10 Pylon (SVT/DVD)	O-Level
Clean Station Clean Station 10 (PATS-70)	O-Level
MAU-50(PATS-70)	I-Level
Stations 3&9 Pylon (with MAU-40 installed)	
Stations 3&9 Pylon (with MAU-40 installed) (SVT/DVD)	O-Level
Clean Station Clean Station 3 (PATS-70)	O-Level
Pylon (PATS-70)	I-Level
Stations 3&9 Pylon (SVT/DVD)	O-Level
Clean Station Clean Station 9 (PATS-70)	O-Level
MAU-40(PATS-70)	I-Level
Stations 4&8 Pylon (with MAU-40 installed)	
Stations 4&8 Pylon (with MAU-40 installed) (SVT/DVD)	O-Level
Clean Station Clean Station 4 (PATS-70)	O-Level
Pylon (PATS-70)	I-Level
Stations 4&8 Pylon (SVT/DVD)	O-Level
Clean Station Clean Station 8 (PATS-70)	O-Level
MAU-40(PATS-70)	I-Level
Station 5 Pylon (with MAU-40 installed)	
Station 5 Pylon (with MAU-40 installed) (SVT/DVD)	O-Level
Clean Station Clean Station 5 (PATS-70)	O-Level
Pylon (PATS-70)	I-Level
Station 5 Pylon (SVT/DVD)	O-Level
MAU-40(PATS-70)	I-Level
Station 6 Pylon (with MAU-40 installed)	
Station 6 Pylon (with MAU-40 installed) (SVT/DVD)	O-Level
Clean Station Clean Station 6 (PATS-70)	O-Level
Pylon (PATS-70)	I-Level
Station 6 Pylon (SVT/DVD)	O-Level
MAU-40(PATS-70)	I-Level
Station 7 Pylon (with MAU-40 installed)	
Station 7 Pylon (with MAU-40 installed) (SVT/DVD)	O-Level
Clean Station Clean Station 7 (PATS-70)	O-Level
Pylon (PATS-70)	I-Level
Station 7 Pylon (SVT/DVD)	O-Level
MAU-40(PATS-70)	I-Level
TER-9/A or Variant installed	
TER-9/A or Variant installed (SVT/DVD)	O-Level
TER-9/A or Variant installed (PATS-70)	I-Level

Common Armament Tester for Fighter (CAT-F)

CAT-F AME Maintenance Level Requirements	
A-10 AME Requirements	MX Level
Dual Rail Adapter (DRA) with	
Dual Rail Adapter (DRA) with LAU-105s installed or LAU-128s installed (SVT/DVD)	O-Level
Dual Rail Adapter (DRA) with LAU-105s installed or LAU-128s installed(PATS-70)	I-Level
LAU-105s installed or LAU-128s installed	
Dual Rail Adapter (DRA) (SVT/DVD)	O-Level
Dual Rail Adapter (DRA)(PATS-70)	I-Level
LAU-131	
LAU-131 Missile Launcher (SVT/DVD)	O-Level
LAU-131 Missile Launcher(PATS-70)	I-Level
Chaff Flare	
Chaff Flare (AN/ALE-47 CMDS) (SVT/DVD)	O-Level
LAU-117	
LAU-117 (SVT/DVD)	O-Level
LAU-117(PATS-70)	I-Level
BRU-61	
BRU-61 (SVT/DVD)	O-Level
BRU-61(PATS-70)	I-Level
LAU-88	
LAU-88 (SVT/DVD)	O-Level
LAU-88(PATS-70)	I-Level
Gun System	
Clean Station Gun System (PATS-70)	O-Level

ATTACHMENT: 4
Common Armament Tester for Fighter (CAT-F)

CAPABILITY DEVELOPMENT DOCUMENT (CDD) SUMMARY

Prepared by:
ATS Engineering Office
AFMC AFLCMC/WNA ATS
Program Office Robins AFB,
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15 July 2024

The United States Air Force (USAF) requires testing capabilities for fighter Aircraft's weapons systems to include legacy and smart weapons. The Program's Capability Development Document (CDD) documented the requirements for developing a common armaments tester. The developed Common Armament Tester for Fighter (CAT-F) will support the F-16, F-15, and A-1) Aircraft, but will also have modular elements which will allow modification to be used on future aircraft.

Key Performance Parameters (KPPs)

The Government Team examined performance attributes which were determined to be significant in the development of CAT-F. The six Key Performance Parameters (KPPs) identified as vital system capabilities are required to meet the operational goals set by the fighter aircraft community. These KPPs are the foundation of the CAT-F development and considered to be essential to performance, function, design, and operations. Any changes to the KPPs would have significant impact on the performance of the CAT-F along with the development, cost, and schedule.

Legacy Testing Requirements/Nuclear Certification

The CAT-F will be capable of performing all current legacy armament system tests and meet required nuclear certification. The CAT-F will replace multiple current USAF fighter aircraft armament testers and shall execute all legacy inspection and troubleshooting requirements. The tester will be required to conduct armament preload and fault isolation testing on active model versions of the F-16 and F-15 fleets. This tester should be considered as a replacement armament tester for the A-10 fleet and be a future consideration for newer generation fighters. To verify the requirement, the CAT-F will be tested in all operationally representative environments and on all appropriate fighter aircraft. The results will be compared to the appropriate legacy tester.

Tester Circuit Protection

The CAT-F tester will be developed with built in Circuit protection. The design will ensure the users, CAT-F, and Unit Under Test (UUT) are protected from improper voltage or amperage. Circuit Protection can be obtained by using a traditional fuse designed to 'blow' when a circuit is overloaded and easily replaced or by incorporating a circuit breaker that features a switch designed to trip by improper conditions. The switch then can simply be flip back into the 'on' position to restore power to the tester, providing no additional repairs are required. Verification of KPP2 will be accomplished by applying enough input power to blow fuse or trip circuit breaker in the designed tester, followed by replacing blown fuse or resetting breaker and verify the tester circuits were not damaged. The appropriate sections of MIL-STD-704 (Aircraft Electrical Power Characteristics) will be identified to ensure power requirements for circuit protection are followed.

Handheld Tester Power

The CAT-F Handheld Tester will be powered by common design batteries which are nonproprietary, non-rechargeable and replaceable in minimal amount of time. Preload testing will utilize battery power for all go/no-go preload and stray voltage testing. In addition to having the required battery power, user should have the option to use Aircraft or Aerospace Ground Equipment (AGE) power for preload testing. Power adapter cables must be provided for all user power options. Verification of handheld preload testing can be accomplished without the use of external power (AGE, Aircraft, building etc.). Verification team will ensure the handheld tester internal batteries can be replaced in under 2 minutes. If required, the verification team will ensure preload testing can easily utilize AGE, Aircraft, and backshop building power to execute requirements.

Ruggedized Common CAT-F Tester

The developed CAT-F will be designed to integrate survivability fundamentals to function reliably in harsh environments such as extremes of temperature, shock and vibration, humidity, or radiation. All CAT-F O-Level/flightline components (Handheld Tester, Preload Kit, Cables...) will meet MIL-PRF-28800G requirements for tailored Class I. The fault isolation kit will meet MIL-PRF-28800G requirements for a tailored Class II. Verified by digital twin representation and/or tester simulation in required operational conditions.

Future Armament Testing Requirements

The CAT-F design will include the capability to upgrade by using the Computer Program Identification Number (CPIN) management process to sustain tester upgrades and modification during the tester's life cycle. The developed CAT-F will be built using an open system software architecture that allows for future upgrades and modifications. In addition to software upgrades, the CAT-F should utilize as much as possible Modular Open Systems Approach (MOSA) design for tester hardware and sustainment. Vendor will be required to submit all requested data to gain government engineering acceptance/approval for the MOSA standard and upgrade capability for software and hardware.

Common Handheld Tester Weight

The CAT-F Handheld Tester shall be built with a light-weight handheld design and include a strap/strain relief (lanyard) which allows the handheld tester to be held or hung during extended use. Handheld tester shall not exceed more than 6 pounds with the batteries installed. The design must include strap/strain relief. The desired weight of the handheld tester is less than 3.5 pounds including batteries. The Handheld tester will be weighed to ensure that it is within specifications.

Mandatory Key Performance Parameters***Sustainment and System Survivability***

The CAT-F shall be structured to ensure testing, operational testing, and system simulation to generate appropriate data to evaluate the fighter's armament system. Tester should be designed to meet or exceed the testing requirements and capabilities of the identified legacy units. The new developed testers shall be able to perform at a minimum all functional and diagnostic testing of the armament systems and test equipment on A-10, F- 15, and F-16 Aircraft. The tester designed shall have the functionality for software updates via Computer Program Identification Number (CPIN) to accommodate all future munitions and testing Aircraft requirements. The Tester shall pass Nuclear Certification for Electrical Design requirements and Nuclear Safety standards. To sustain the CAT-F effort, the design shall be supported by life-cycle procurement resources to replace and repair malfunctioning testers. Tester shall be developed with a Modular Open Systems Approach (MOSA) and defined as having a technical and business strategy for designing an affordable and adaptable system. Tester development shall produce research to determine the baseline of the existing legacy testers' supportability posture, to include

Reliability, Maintainability, Diminishing Manufacturing Sources and Material Shortages (DMSMS), and Part Obsolescence. The Government Reference Architecture (GRA) strategy will define actions required to maintain the testers over the established life cycle. The CAT-F should be designed to function as intended in operating locations. The armament tester must meet all required environmental requirements. Tester shall incorporate the required ruggedized standards to be utilized at both O-Level and I-Level maintenance. Tester shall withstand all vibration, and explosive atmosphere testing in accordance with MIL-PRF-28800. The Sustainment and System Survivability Mandatory Parameters are covered in the 6 KPPs.

Key System Attribute (KSA)

The CAT-F Joint IPT and Key Stakeholder Team examined performance attributes considered important but not critical to meeting system goals. These attributes were classified as a Key System Attribute (KSA). Listed in Table 11-1 are the twelve KSAs identified as required to achieve needed system capabilities to meet the operational goals set by the fighter aircraft community. These KSAs are also part of the foundation of the CAT-F development and provide decision-makers with an additional level of capability performance characteristics below the KPP level. Advances in technology or changes to system goals may result in changes to the KSA's threshold and objective

TPS Kit Weight

The weight of each TPS kit shall not exceed 79 pounds per case inclusive of all cables, connectors, adapters, etc. that are contained within. The preferred Kit weight is less than 44 pounds using a dispatchable case with handles and wheels.

Fault Isolation Tester Power

This equipment shall utilize electric power in accordance with MIL-STD-704 and shall be in accordance with the utilization equipment requirements of MIL-STD-704. As a minimum the primary power source shall operate at 100-240V@400 Hz and 28VDC in accordance with MIL-STD-704 and include adapters to utilize both Aircraft, AGE, and backshop power.

Electronic Storage

Tester will be required to create and store testing results. The CAT-F shall create log files for all testing. The CAT-F must have ability to export test report log files to another computing device via direct connect. Tester will have minimum of 256 GB storage, able to store 7 days of data, and export log files. Greater than 256 GB storage and able to store 30+ days of data, and export log files is preferred.

Built-In-Test (BIT)

Tester will integrate BIT capable of failure detection, fault isolation, monitoring, and provide indications of out-of-range tester and cable conditions. The equipment shall be able to provide and store automatic diagnostic self-test information. At a minimum, the information provided shall indicate whether the equipment is functional and identify malfunctions. If feasible, identify malfunctions with fault isolation information.

Tester Repairability

Tester should be modular design to allow unit repairability to maximum extent possible. Cables can be repaired at the field level to the maximum extent possible using current AF standard tools without vendor proprietary tooling. Tester components will be 80% repairable or replaceable at the unit level. All TPS cables will be equal or greater than 80% repairable and the cable connector assemblies must be fully repairable at base level. The design of the equipment shall permit isolation of faults and repair down to the component item or lowest replaceable unit with the maintenance provisions furnished.

Tester Dimensions (Handheld)

The CAT-F handheld Tester Dimensions shall accommodate required manual controls and readable screen. The total volume should be no more than 240 cubic inches (example L:12, H:5, and W:4 inches). Design shall be easily held in one hand by the user during operation.

Test Duration

Total test time using the Common Handheld Tester and the Preload TPS/Kit (O-level MDS specific) shall be less than the current duration of all comparable operational and/or functional testing. The O-LVL 180-day testing time will not exceed 6 Hours and all testing at I-LVL will be under 1 hour per AME.

Test/Tester Set-up

The CAT-F will be designed to reduce the number of connect / disconnect test steps for a single piece of armament equipment and allow for testing of multiple pieces of equipment simultaneously. The CAT-F tester set-up time should not be more than 50% of testing time.

I-LVL & O-LVL Cable Consolidation

Backshop (I-LVL) and Flightline (O-LVL) Cables should incorporate standard cables for the same or equivalent UUT. I-LVL & O-LVL cable kits will be interchangeable and be a common configuration. If possible, the CAT-F tester would also be able to utilize legacy cables.

Tester Calibration

The CAT-F will be designed to reduce downtime by eliminating the need for calibration or maximizing the time period between calibrations. Air Force Metrology and Calibration (AFMETCAL) initial certification calibration interval shall be longer than 12 months, but no calibration is preferred.

Common Handheld Tester Display Size

Common Handheld Tester Display Size shall depend on final tester dimensions. Screen should follow current military standards with sunlight readable graphic display with resolution at least 128 X 64 pixels or better, and different screen background. The handheld tester display will be at a minimum 1.5 inches by 2.5 inches and display output shall be distinguishable to operator in all required environments.

Tester Reliability

The CAT-F design will maintain an operational reliability in accordance with Military Standard MIL-PRF-28800G. Equipment shall be such that under normal use and operation the equipment does not fail within 1500 hours of operation with a statistical certainty of 95%. Reliability tests shall be performed by evaluation of manufacturer's data to establish that reliability falls within the parameters outlined.

Requirements, Thresholds, and Objectives

The Government Team developed the KPP and KSAs to ensure early sustainment planning and enabling the requirements and acquisition communities to provide a capability solution with optimal availability, maintainability, and reliability to the warfighter at an affordable life cycle cost. CAT-F System Requirements below lists the key KPPs and KSA with required thresholds as well as objectives.

Type	Attribute	Requirement	Threshold	Objective
KPP	Legacy Testing Requirements/ Nuclear Certification	Common Armament Tester will be capable of performing all current legacy armament system tests and meet required nuclear certification.	Replace F-15 and F-16 USAF fighter Aircraft armament testers.	Replace F-16, F-15, and A-10 USAF fighter Aircraft armament testers.
KPP	Tester Circuit Protection	Circuit protection shall ensure the tester is not damaged from input power or the Unit Under Test (UUT) and shall ensure the UUT is not damaged from input power or the tester. (MIL-HDBK-704)	Circuit protection: Integrate a limiting fuse to ensure that neither the UUT or the tester is damaged from input power or each other.	Circuit protection: Integrate a circuit breaker to ensure that neither the UUT or the tester is damaged from input power or each other.
KPP	Handheld Tester Power	Handheld Tester will be powered by common design batteries which are non-proprietary, non-rechargeable and replaceable in minimal amount of time.	Preload testing will utilize battery power for all go/no-go preload and stray voltage testing.	In addition to threshold, will have the option to use the fighter Aircraft/AGE power for preload testing (All power adapter cables must be provided).
KPP	Ruggedized Common CAT-F Tester	Developed armament tester will be designed to integrate survivability fundamentals to function reliably in harsh environments such as extremes of temperature, shock and vibration, humidity, or radiation.	Handheld Tester, Pre-Load Kit, and all cables will meet MIL-PRF-28800G requirements for tailored Class I. Fault isolation kit will meet MIL-PRF-28800G requirements for tailored Class II.	All Tester elements will meet MIL-PRF-28800G requirements for Class I or tailored Class I.
KPP	Future Armament Testing Requirements	Tester designed will include the capability to upgrade SOFTWARE by using the Computer Program Identification Number (CPIN) management process to sustain tester upgrades and modification during life cycle.	Built using an open system software architecture that allows upgrade and modification capability by utilizing the Computer Program Identification Number (CPIN) process.	In addition to threshold, utilize a modular open systems approach for tester hardware.

Type	Attribute	Requirement	Threshold	Objective
KPP	Common Handheld Tester Weight	Handheld Tester Weight shall be built with a light-weight handheld design and include a strap/strain relief (lanyard) which allows the handheld tester to be held or hung during use.	Handheld Tester cannot exceed 6 pounds with batteries installed. Design must include strap/strain relief.	Handheld Tester will weigh less than 3.5 pounds including batteries. Design must include strap/strain relief.
KSA	TPS Kit Weight	The weight of each TPS kit will not exceed 79 pounds per case inclusive of all cables, connectors, adapters, etc. that are contained within.	The TPSs weight will not exceed 79 pounds per case.	The TPSs weight will be less than 44 pounds using dispatchable case with handle and wheels.
KSA	Fault Isolation Tester Power	This equipment shall utilize electric power in accordance with MIL-STD-704 and shall be in accordance with the utilization equipment requirements of MIL-STD-704.	Primary power source shall operate at 100-240V@400 Hz and 28VDC IAW MIL-STD-704 and include adapters to utilize both aircraft, AGE, and backshop power.	Tester shall also be able to utilize 100-240V@50/60Hz with auto-ranging capability, in addition to threshold IAW MIL-STD-704.
KSA	Electronic Storage	Handheld Tester will store test report log files created during O-LVL, I-LVL, BIT, and self-testing. Tester must have ability to export test report log files to another computing device via direct connect.	Tester will have minimum of 256 GB storage, able to store 7 days of data, and export log files.	Greater than 256 GB storage and able to store 30+ days of data.
KSA	Built-In-Test (BIT)	Tester will integrate BIT capable of failure detection, fault isolation, monitoring, and provide indications of out-of-range tester and cable conditions.	The equipment shall be able to provide automatic diagnostic self-test information. At a minimum, the information provided shall indicate whether the equipment is functional or identify malfunctions and store non-classified data.	Data shall indicate whether the equipment is functional and identify malfunctions with fault isolation information.
KSA	Tester Repairability	Tester should be modular design to allow unit repairability to maximum extent possible. Cables can be repaired at the field level to the maximum extent possible using current AF standard tools without vendor proprietary tooling.	Tester: 80% of all components will be repairable/replaceable at the unit level. Cables: 80% of all cables in the TPS will be repairable. Cable connector assemblies will be fully repairable at base level.	Tester: 100% of all components will be repairable/replaceable at the unit level. Cables: 100% of all cables in the TPS will be repairable. Cable connector assemblies will be fully repairable at base level.
KSA	Tester Dimensions (Handheld)	Tester Dimensions - Shall accommodate any required manual controls and a readable screen.	The total volume should be no more than 240 cubic inches (example L:12, H:5, and W:4 inches). Design shall be easily held in one hand by the user during operation.	Smaller than the threshold.
KSA	Test Duration	Test Duration - Total test time using the Common Handheld Tester and the Preload TPS/Kit (O-level MDS specific) shall be less than the duration of the operational and/or functional testing.	O-LVL 180-day testing time will not exceed 6 hours and all testing at I-LVL will be under 1 hour per AME.	O-LVL 180-day testing time will not exceed 4.5 hours and all testing at I-LVL will be under 45 minutes per AME.

Type	Attribute	Requirement	Threshold	Objective
KSA	Test/Tester Set- up	Reduce the number of connect / disconnect test steps for a single piece of armament equipment and allow for testing of multiple pieces of equipment simultaneously.	Set-up time should not be more than 50% of testing time.	Smaller than the threshold.
KSA	I-LVL & O- LVL Cable Consolidation	I-LVL & O-LVL Cable Consolidation - Standard I- LVL & O-LVL cables for the same or equivalent UUT.	Tester cable kits will be interchangeable for I- LVL & O-LVL testing.	In addition to threshold, tester will utilize legacy cables when practical.
KSA	Tester Calibration	Tester Calibration - Minimize calibration downtime.	Calibration interval shall be longer than 12 months.	Test would not require Calibration.
KSA	Common Handheld Tester Display Size	Common Handheld Tester Display Size - Size of display will depend on final tester dimensions. Screen should follow current military standards with sunlight readable graphic display with resolution at least 128 X 64 pixels or better, and different screen background.	The handheld tester display will be at a minimum 1.5 inches by 2.5 inches with resolution at least 128 X 64 pixels and display output shall be distinguishable to operator in all required environments.	Handheld tester display will be larger than 1.5 inches by 2.5 inches while meeting threshold requirement.
KSA	Tester Reliability	Tester design to maintain operational reliability in accordance with Military Standards.	Design of equipment shall be such that under normal use and operation the equipment does not fail within 1500 hours of operation with a statistical certainty of 95%.	Design of equipment shall be such that under normal use and operation the equipment does not fail within >1500 hours of operation with a statistical certainty of 95+%.

PREPARATION OF THE STATEMENT OF WORK (SOW)

Notional example of CONTRACTOR STATEMENT OF WORK.

1. SCOPE

1.1 Background (Optional)

2. APPLICABLE DOCUMENTS

2.1 Applicable documents. All documents invoked in the requirements section of the SOW must be listed in this section by document number, title, date or by revision letter. These documents may include Standards, Specifications and other reference documents needed to identify and clarify the work task or deliverable product. DoD and Departmental Instructions should not be used in the SOW to control contractor effort. Also, any document listed in this section must be invoked and selectively tailored to meet minimal needs of the planned procurement in the requirements section. The version of any document cited in the SOW should be specified in this section.

2.2 Department of Defense specifications.

2.3 Department of Defense standards.

2.4 Other Government documents.

2.5 Industry documents.

3. GENERAL REQUIREMENTS (FUNDING TYPE)

3.1 General requirements. (For guidance only)The content of the SOW is dependent on which phase of the acquisition cycle you are in. A SOW for a Research & Development effort is different than production. Remember work requirements are stated in the SOW, not how to build it. The funding type identified at the “GENERAL REQUIREMENTS” paragraph implies one funding type to support the effort.

3.2 Program Management. The contractor shall prepare progress, status and management reports. The reports address the progress of the effort, status of any problem resolution or identification of new problems, summary of safety-related actions, significant events, and updated contract schedule.

a. CDRL Deliverable(s):

b. Progress, Status, and Management Report

3.3 Program Reviews. The contractor shall conduct quarterly reviews, report on cost, schedule, technical progress, program risks and mitigations, and technical performance

measures. The contractor shall present the Risk Management Database and Risk Management Chart as part of the Project Management Review Presentation Materials.

- a. CDRL Deliverable(s)
- b. Conference Agenda
- c. Presentation Material
- d. Report, Record of Meeting / Minutes

3.4 System Engineering (SE). The contractor shall ensure the effective development and delivery of capability through the implementation of a balanced approach with respect to cost, schedule, performance, and risk using integrated, disciplined, and consistent SE activities and processes. SE-related policy, guidance, specifications, and standards are intended to successfully guide the technical planning and execution of a program across the acquisition life cycle.

3.5 Environment, Safety, and Occupational Health (ESOH). The contractor shall perform ESOH hazard identification and risk management in accordance with MIL-STD-882. The safety assessment identifies safety-related hazards that may be present in the physical system configuration, including specific procedural controls and precautions that should be followed for risk mitigation. Identify hazardous materials contained in or used with the system (Note a CDRL should be included which will include required content). Include environmental and occupational health risks in ESOH assessments. The contractor shall identify hazardous materials for systems per DoDI 5000.85 and DoDI 5000.88.

3.6 Reliability and Maintainability Program. The contractor shall create and maintain a Failure Reporting and Corrective Action System (FRACAS) to document all problems found during the manufacturing, fabrication, assembly, test, operation, and maintenance aspects of the system life cycle for each system produced.

- a. CDRL Deliverable(s):
- b. FRACAS Report

3.7 Data. The contractor shall include technical data and rights to that data which are described in relation to the technical design and implementation as well as cost, schedule, performance, and risk in this section. Data related topics are described such as:

- a. Compatibility requirements with DoD computing infrastructure.
- b. Compatibility requirements with DoD physical infrastructure.
- c. Description of drawings, practices, procedures and deliverables.
- d. Hardware and software information

3.8 Human System Integration (HSI). As part of SE, the contractor shall integrate consideration for the human part of the system into the design and operations of the

system by inserting into manpower, personnel, training, human factors engineering, environment, safety, occupational health hazards, and personnel survivability areas. (Describe the Manpower, Personnel and Training and Education requirements for a cybersecurity workforce needed in the Cybersecurity section 3.11)

3.9 Electromagnetic (EM) Spectrum Requirements and Electromagnetic Environmental Effects (E3). For spectrum dependent systems, the contractor shall identify what frequency assignments and spectrum certifications such as DD Form 1494 are needed. E3 control is concerned with proper design and engineering to minimize the impact of the EM environment on equipment, systems, and platforms. See standards and handbooks such as MIL-STD-461 in ASSIST.

3.10 Technical Reviews and Updates. The contractor shall provide SE and technical reviews that include technical and programmatic cost, schedule and performance status and risks. These reviews shall be in coordination with other required reviews and meetings in the SOW to increase coordination and reduce duplication.

- a. CDRL Deliverable(s)
- b. Conference Agenda.
- c. Presentation Material.
- d. Report, Record of Meeting / Minutes.

3.11 Cyber Security. The contractor shall ensure procedures and processes are in place for the protection of DoD program information residing on or transiting DoD and corporate networks and information systems. The objective is to protect DoD information, not just DoD systems, and it relates to all programs, not just those Information Technology (IT) focused.

Address cybersecurity workforce challenges and efforts designed to improve Manpower, Personnel, Training and Education in a challenging cyber environment. Describe the type of workforce available to work on cyber security issues.

3.12 Configuration Management. The contractor shall document life cycle CM planning and provide CM program milestones and schedules to monitor program status. Implement an internal CM system for the control of all configuration documentation, physical media, and physical parts. For software, the system will address the evolving configuration and support environments (engineering, implementation, and test) used to generate and test the product.

3.13 Configuration Change. The contractor shall establish and maintain a Configuration Status Accounting (CSA) database. All baselines and changes must be documented in the database. Provide CSA reporting in accordance with the guidelines in section 7 "Configuration Status Accounting" in MIL-HDBK-61. Provide the traceability of all changes

from the original baseline configuration documentation of each configuration item. (Note this content will be listed in a CDRL/DID)

a. CDRL Deliverable(s):

b. Configuration Status Accounting Report

3.14 Packaging Handling, Storage and Transportation (PHST). The contractor shall package and handle hardware and related parts in accordance with MIL-STD-2073-1 DoD Standard Practice for Military Packaging.

3.15 Diminishing Manufacturing Sources and Material Shortage (DMSMS). The contractor shall manage obsolescence over the entire period of the contract, and notwithstanding any obsolescence issues or problems, the contractor remains responsible for meeting all performance and other requirements of this contract.

a. CDRL Deliverable(s):

b. Annual Obsolescence Report

3.16 Physical Configuration Audit. The contractor shall conduct a Physical Configuration Audit (PCA) jointly with the Government, as well as develop and deliver a Configuration Audit Plan. (CDRLA002) At the completion of PCA and all action items are closed, the contractor will develop and deliver a Configuration Audit Summary Report. (CDRLA018)

a. CDRL Deliverable(s):

b. Configuration Audit Summary Report (Physical)

c. Configuration Audit Plan

3.17 Provisioning Items Ordering Spares. The contractor shall deliver the spare parts for provisioning orders. Provisioning Items Ordering (PIO) will be ordered by the Government in accordance with the PIO CLIN in the contract which is defined upon contract award.

(Note If cited elsewhere in the RFP/contract, should not be repeated in the SOW)

3.18 Reliability and Maintainability Program. The contractor shall create and maintain a Failure Reporting and Corrective Action System (FRACAS) to document all problems found during the manufacturing, fabrication, assembly, test, operation, and maintenance aspects of the system life cycle for each system produced.

a. CDRL Deliverable(s):

b. FRACAS Report

3.19 Test and Certification. The contractor shall submit a Production Test Plan, which will address Low-Rate Initial Production (LRIP) and Full Rate Production (FRP) test events.

a. CDRL Deliverable(s):

b. Production Test Plan

4. NOTES

4.1 Definitions. Provide definitions of key terms such as: Request for Variance (RFV) - Is the term for what is formerly known as a Request for Deviation (RFD) or an RFW. A RFV enables the Government to determine the impact on performance, operational readiness, logistics support or other affected areas.

Government areas of interest with the CAT-F Acquisition SOW.

1. SCOPE

- a. Program Synopsis
- b. Overview of the CAT-F Program for fighter aircraft F-16, F-15, and A-10
- c. Government Data Rights

2. APPLICABLE DOCUMENTS

3. GENERAL REQUIREMENTS

- a. PROGRAM CONFERENCES
 - Post-Award Conference (PAC)
 - Engineering Data Guidance Conference
 - Technical Order (TO) Guidance Conference
 - Provisioning Guidance Conference
- b. GENERAL TECHNICAL REQUIREMENTS
 - General Design Requirements
 - Technical Approach
 - TPS General Design Requirements
 - CAT-F System Software Requirements
- c. NUCLEAR CERTIFICATION
 - Nuclear Certification Impact Statement (NCIS)
 - Certification Requirements Plan (CRP)
 - Nuclear Safety Analysis Report (NSAR)
- d. RELIABILITY
- e. PROGRAM PROTECTION
 - Program Protection Implementation Plan (PPIP)
 - Inspection and Audit
 - Cybersecurity
 - Anti-Tamper (AT)
 - Counterfeit Prevention
- f. SAFETY
 - Hazard Analysis and Risk Mitigation
 - Environmental, Safety, and Occupational Health (ESOH)
- g. SYSTEMS ENGINEERING TECHNICAL REVIEWS (SETRS)
- h. PROGRAM MANAGER (PM)

- Program Management Activities
- Earned Value Management (EVM)
- Program Reviews and Conferences
- i. LOGISTICS
 - Acquired or Furnished Property
 - Technical Orders (TOs)
 - Engineering Data
 - Familiarization
 - Provisioning
 - Configuration Management (CM) and Control
 - Diminishing Manufacturing Sources and Material Shortages (DMSMS)
 - Parts Control and Standardization Program
 - Data Management
- j. DATA ACCESSION LIST (DAL)
- k. SECURITY
 - Security Information
 - Contractor Facilities
 - Security Information
 - Contractor Facilities
 - Communications Security (COMSEC)
- l. ENGINEERING AND MANUFACTURING DEVELOPMENT (EMD) PHASE
 - Develop/Build Production Representative Article TPSs
 - CAT-F Preliminary Design Review (PDR)
 - CAT-F System Critical Design Review (CDR)
 - CAT-F System Verification Review (SVR)
 - CAT-F Calibration and Measurement Requirements Summary (CMRS)
 - Environmental Testing
 - Cybersecurity Testing
 - UUT Test Requirements Documents (TRDs)
 - Operational Test Readiness Review (OTRR)
 - Operational Test and Evaluation (OT&E) Support
 - CAT-F O-Level Handheld Tester Test Requirements Documents (TRDs)
 - CAT-F I-Level F/I Tester Test Requirements Documents (TRDs)
 - Production Readiness Review (PRR)
- m. PRODUCTION
 - CAT-F Software TPS Production
 - CAT-F Cables and Adapters
 - CAT-F O-Level Handheld Tester Production
 - CAT-F I-Level Fault Isolation Tester Production
- n. WARRANTY